

Scalable Variational Inference for Boolean Matrix Factorization with an Unknown Latent Dimension

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Abstract

Boolean matrix factorization (BMF) aims to factorize a binary matrix into an approximate Boolean product of two smaller binary factor matrices. Its flexible nonlinear nature and nice interpretability lends itself to many applications in scientific disciplines and machine learning. We focus on a popular probabilistic version of BMF – the deterministic input, noisy output *and* gate (DINA) model – a discrete latent variable model widely used in educational measurement for cognitive diagnosis of students. The higher-order interaction effects induced by the Boolean product pose significant statistical and computational challenges, especially when the underlying latent dimension is unknown and the observed data dimension is large. We propose a novel variational inference method for estimating the parameters of a probabilistic BMF as well as the unknown latent dimension. Our mean-field variational inference approach incorporates a nonparametric cumulative shrinkage process prior and stochastic coordinate ascent updates. This approach allows scaling the estimation of the latent dimension and the binary factor matrices to big datasets with minimal hyperparameter tuning. Extensive simulation studies demonstrate the excellent empirical performance of our method. We apply the method to real datasets from educational assessments and single-cell genomics and obtain interpretable findings.

Keywords: Cognitive diagnosis model; Cumulative shrinkage process; Boolean matrix factorization; Discrete latent variable model; Gradient estimation; Variational inference; Non-parametric Bayes.

1 Introduction

Boolean matrix factorization (BMF) aims to factorize a binary matrix into an approximate Boolean product of two smaller binary factor matrices. Its flexible nonlinear nature and nice interpretability lends itself to many applications in various fields. On the one hand, BMF is

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a popular dimension reduction and pattern recognition paradigm in machine learning (Miettinen and Vreeken, 2011; Rukat et al., 2017a; Wan et al., 2020; Miettinen and Neumann, 2021). On the other hand, thanks to its easily interpretable nature, BMF has also received considerable interest across scientific disciplines. BMF has been used for studying transcriptional data in bioinformatics (Liang et al., 2020), for identifying functional interactions in brain networks in neuroscience (Haddad et al., 2018), for adding missing attribute information to products in the Amazon catalog (Rukat et al., 2017b), and for modeling conjunctive and disjunctive rules in consumer choice modeling (Dzyabura and Hauser, 2011). In particular, a probabilistic version of BMF – the deterministic input, noisy *and* gate (DINA) model (Junker and Sijtsma, 2001) – is a very popular and fundamental model in psychometrics for performing cognitive diagnosis of students based on their response data in educational assessments. We will use the DINA model as a key motivating example for our contributions.

The DINA model belongs to the big family of Cognitive Diagnostic Models (CDMs or diagnostic classification models; Rupp and Templin, 2008b; von Davier and Lee, 2019), which are discrete latent variable models widely used for modeling multivariate data in educational and psychological assessments. The observed data for DINA or other CDMs are usually N subjects’ multivariate binary responses to a set of P items. The goal of fitting a CDM includes inferring the subjects’ presence or absence of a set of K fine-grained latent attributes, such as cognitive abilities or mental disorders. One key structure in a CDM is the so-called Q -matrix (Tatsuoka, 1983), which describes how the J observed items depend on the K latent attributes. Phrased in the BMF terminology, in the DINA model, the observed binary matrix is the $N \times P$ response matrix collecting N students’ correct/wrong responses to P test questions, and the two binary factor matrices correspond to (a) the $P \times K$ item-attribute matrix (Q -matrix) and (b) the $N \times K$ attribute mastery matrix collecting the N students’ latent mastery/deficiency statuses of the K attributes. As a probabilistic version of BMF, the DINA model assumes a conjunctive Boolean relationship among attributes and

incorporates additional continuous parameters (guessing and slipping probabilities) to model the item responses. Identifiability and estimation of the DINA model has received substantial interest (see, e.g. [Rupp and Templin, 2008a](#); [de la Torre, 2009](#); [DeCarlo, 2011](#); [Culpepper, 2015](#); [Chen et al., 2018](#); [Yamaguchi and Okada, 2020b](#)), but the significant computational and statistical challenges still remain to be addressed, especially for high-dimensional scenarios.

There are several critical limitations of existing estimation methods for the DINA model and probabilistic BMF in general. [YG: I will add a few sentences on other non-VI methods for DINA.] Recently, variational inference (VI) has been employed in the inference of CDMs including the DINA model ([Hijikata et al., 2023](#)). However, existing VI approaches face significant scalability limitations. [Yamaguchi and Okada \(2020b\)](#) reformulated the DINA model as a mixture model by assigning each possible attribute mastery pattern of a respondent as a latent class, necessitating the estimation of $N \times 2^K$ latent class variables. Similarly, [Oka and Okada \(2023\)](#) built on this latent class model and proposes an iterative variational method that evaluates all possible attribute patterns for each item and selects the item-attribute matrix with the highest posterior probability, requiring $J \times (2^K - 1)$ posterior evaluations per iteration. Consequently, the feasible number of attributes for these methods remains relatively small, typically with $K \leq 8$ ([Yamaguchi and Okada, 2020b](#); [Oka and Okada, 2023](#)). Subsequent work generalizes this variational Bayes method to the multiple-choice DINA model ([Yamaguchi, 2020](#)). In the CDM literature, many existing methods have been limited in their application to relatively small datasets or otherwise require knowledge of an expert-derived item-attribute matrix (i.e., the Q-matrix). In the BMF notation, this means one of the two binary factor matrices is already known. Furthermore, most methods assume knowledge of the latent dimension K . In the literature, there have been some recent works on variational inference for cognitive diagnostic models, such as [Yamaguchi and Okada \(2020b\)](#) and [Yamaguchi and Okada \(2020a\)](#). However, none of these existing works can deal with an unknown number of latent attributes. Scalable estimation methods would

allow the DINA model to be applied in new contexts, where the number of latent attributes and the number of items may be very large; for example, outside of the education context, data resulting from biological sequencing experiments where hundreds to thousands of genes or chromosomal regions are profiled per experiment. Here, we propose an approach based on variational inference that accomplishes this goal.

[YG: some of the following discussions on VI seem to be very advanced and not directly related to our approach; maybe move some to the later algorithm section and be more concise here?] Variational inference for discrete latent variables has recently drawn significant interest in machine learning and statistics. The main challenges in developing fast and accurate variational methods are the non-differentiability of discrete variables and the combinatorial problem in high-dimensional spaces. Most work deals with these challenges by designing stochastic gradient descent methods (Robbins and Monro, 1951). One approach is to apply continuous approximation to discrete variables, which aims to trade bias for low variance. For example, the Gumbel-Softmax technique approximates sampling from a categorical distribution by creating a differentiable relaxation using the Gumbel distribution (Jang et al., 2016; Maddison et al., 2016). The other approach is built on the gradient estimation using the score function method (Mohamed et al., 2020). The score function gradient estimation is unbiased but often has high variance, which requires variance reduction methods to be practically useful. Some methods propose control variates using Taylor expansions (Titsias and Shi, 2022), neural networks (Tucker et al., 2017), and Stein operators (Shi et al., 2022). Other methods develop antithetic sampling techniques based on the symmetry of the sampling space (Yin and Zhou, 2019; Kunes et al., 2023), and use Rao-Blackwellization with partial integration (Titsias and Lázaro-Gredilla, 2015; Yin et al., 2020; Dong et al., 2020).

Our Contributions. Methodologically, we propose an efficient variational inference algorithm for the DINA model and probabilistic BMF by carefully exploiting the nonlinear interaction-effect-based structure of the likelihood. Our mean-field variational inference ap-

proach uses a nonparametric cumulative shrinkage process prior and stochastic coordinate ascent updates. This approach drastically improves the scalability (e.g., from $K \leq 8$ to $K > 30$) in estimating binary factor matrices to big datasets. In addition, our method is able to automatically infer the latent dimension with minimal hyperparameter tuning. Theoretically, we develop theory for our variational inference algorithm by establishing the sample complexity at which the true parameters become fixed points of the coordinate ascent procedure. We then determine the minimax lower bound, which is equal to the sample complexity of our approach up to logarithmic factors. In terms of applications, we apply the method to two very different real-world datasets, one from international educational assessments and the other from single-cell genomics, and obtain interpretable substantive findings.

The rest of this manuscript is organized as follows. Section ...

2 Model Setup and Background

2.1 Deterministic Input, Noisy output And gate (DINA) Model

We first introduce some notation. For a matrix $\mathbf{M}$, denote its i th row by $\mathbf{M}_{i,:}$. For two vectors $\mathbf{a} = (a_1, \dots, a_L)$ and $\mathbf{b} = (b_1, \dots, b_L)$ of the same length, we write $\mathbf{a} \succeq \mathbf{b}$ if $a_l \geq b_l$ for all $l \in [L]$.

The DINA model is a very popular and fundamental cognitive diagnostic model proposed by [Junker and Sijtsma \(2001\)](#) that uses the Boolean “and” product to model the conjunctive relationship of latent cognitive abilities. Consider N students that respond to P test questions, then the observed data is an $N \times P$ binary matrix $\mathbf{Y} = (Y_{ij}) \in \{0, 1\}^{N \times P}$, where $Y_{ij} = 1$ or 0 indicates the i th student answers the j th question correctly or incorrectly. The two binary factor matrices are: (a) the $P \times K$ item-attribute matrix (Q -matrix) and (b) the $N \times K$ matrix collecting the N students’ latent mastery/deficiency statuses of the K attributes. Many typical matrix factorizations are based on the inner products and often

have the latent factors additively contribute to the expected value of a response variable [YG: add some references]. In contrast, the latent attributes in the DINA model *interact* to determine the response variable's expected value. The *ideal response* of subject i to item (feature) j is defined as

$$\mathbb{1}(\mathbf{A}_{i,:} \succeq \mathbf{G}_{j,:}) = \mathbb{1}(A_{ik} \geq G_{jk} \text{ for all } k \in [K]) = \prod_{k=1}^K A_{ik}^{G_{jk}}.$$

The ideal response equals 1 if and only if $A_{ik} = 1$ whenever $G_{jk} = 1$ for all k , meaning all the required components are equipped by the respondent i . When the ideal response is 1, the respondent has a high probability $1 - s_j$ of success, described by the *slipping parameter* $s_j < 0.5$ corresponding to question difficulty. When $\prod_{k=1}^K A_{ik}^{G_{jk}} = 0$, the respondent is more often wrong with probability $1 - g_j$, and the distribution is described by the *guessing parameter* $g_j < 0.5$. The model can hence be written as

$$\mathbb{P}(Y_{ij} = 1) = \left(\prod_{k=1}^K A_{ik}^{G_{jk}} \right) (1 - s_j) + \left(1 - \left(\prod_{k=1}^K A_{ik}^{G_{jk}} \right) \right) g_j. \quad (1)$$

Of particular interest is the log-likelihood function, which forms the basis of the optimization objectives considered in this paper. To ease the notational burden we introduce the separate components of the log-likelihood function:

$$\psi_{1,j}(Y_{ij}) := Y_{ij} \log(1 - s_j) + (1 - Y_{ij}) \log(s_j), \quad \psi_{2,j}(Y_{ij}) := Y_{ij} \log(g_j) + (1 - Y_{ij}) \log(1 - g_j).$$

where $\psi_{1,j}(Y_{ij})$ and $\psi_{2,j}(Y_{ij})$ are the log-likelihood of Y_{ij} given the ideal response is 1 and 0, respectively. With this notation, the log-likelihood is defined as:

$$\mathcal{L}(\mathbf{A}, \mathbf{G}) := \sum_{i=1}^N \sum_{j=1}^P \left[\left(\prod_{k=1}^K A_{ik}^{G_{jk}} \right) \psi_{1,j}(Y_{ij}) + \left(1 - \prod_{k=1}^K A_{ik}^{G_{jk}} \right) \psi_{2,j}(Y_{ij}) \right] \quad (2)$$

Previous methods directly maximize the log-likelihood (Gu and Xu, 2023), which has com-

putational and statistical advantages despite being a non-convex optimization problem. The DINA model is closely related to the deterministic input, noisy *or* gate (DINO) model (Templin and Henson, 2006), and techniques developed for estimation in the DINA model context can be straightforwardly extended to DINO. In Section § 7.2, we modify our method to fit the DINO model on single-cell assay for transposase accessible chromatin (ATAC) sequencing data. Despite non-convexity due to the product over latent components, the DINA model log-likelihood is linear in each component when holding the others fixed leading to convenient coordinate-wise optimization approaches.

[YG: Mathematical equivalence between DINA and DINO established in Gu and Xu (2019). Emphasize that our contribution will apply to both models and be of use in various applications including BMF, educational cognitive diagnosis (DINA), and psychiatric evaluation (DINO), biomedical application such as single-cell data dimension reduction (DINO).]

2.2 Variational Inference

In our work, we consider approximate posterior inference under a group sparsity inducing prior. We use variational inference to fit an approximate posterior (Blei et al., 2017). In variational inference, we consider a family of possible approximate posterior distributions $\{Q : Q \in \mathcal{Q}\}$ and select the closest to the true posterior via optimization. For observed data $\mathbf{Y}$ and parameter θ , the object of inference is the posterior distribution $P(\theta|\mathbf{Y}) \propto \mathcal{L}(\theta; \mathbf{Y})p(\theta)$ where $\mathcal{L}(\theta; \mathbf{Y})$ is the likelihood function and $p(\theta)$ is a prior distribution. In order to fit a posterior approximation, variational inference minimizes the KL divergence between an approximating distribution and true posterior, which does not depend on the posterior distribution normalizing constant directly:

$$\hat{Q} = \operatorname{argmin}_{Q \in \mathcal{Q}} D_{\text{KL}}(Q, P(\theta|\mathbf{Y})) \quad (3)$$

Point estimates derived from variational inference have shown to have good performance in practical contexts and theoretical properties (Wang and Blei, 2019). Moreover, variational inference has recently been applied in the context of Bayesian nonparametric models (Moran et al., 2022). Previously, the variational inference was applied to DINA model inference (Oka and Okada, 2023); however, the method requires a full enumeration of 2^K possible latent patterns, prohibiting scaling to large numbers of latent component. Moreover, it requires a known number of components that is often not available in practice.

3 Mean-field Variational Inference for DINA

The mean-field approach allows us to parameterize the 2^K possible latent patterns via K variational parameters. Namely, we approximate the joint posterior $P(\mathbf{A}, \mathbf{G} | \mathbf{Y})$ with the factorial mean-field approximation:

$$q(\mathbf{A}, \mathbf{G}; \alpha, \gamma) := \prod_{i=1}^N \prod_{k=1}^K \alpha_{ik}^{A_{ik}} (1 - \alpha_{ik})^{1-A_{ik}} \prod_{j=1}^P \prod_{k=1}^K \gamma_{jk}^{G_{jk}} (1 - \gamma_{jk})^{1-G_{jk}} \quad (4)$$

This approximate posterior leads to a natural continuous relaxation of the DINA log-likelihood; namely, the expected log-likelihood under the variational distribution Q is given by:

$$\begin{aligned} \mathbb{E}_Q[\mathcal{L}(\mathbf{A}, \mathbf{G})] &= \sum_{i=1}^N \sum_{j=1}^P \mathbb{E}_Q \left[\left(\prod_{k=1}^K A_{ik}^{G_{jk}} \right) \psi_{1,j}(Y_{ij}) + \left(1 - \prod_{k=1}^K A_{ik}^{G_{jk}} \right) \psi_{2,j}(Y_{ij}) \right] \\ &= \sum_{i=1}^N \sum_{j=1}^P \mathbb{E}_Q \left[\left(\prod_{k=1}^K (1 - (1 - A_{ik}) G_{jk}) \right) \psi_{1,j}(Y_{ij}) + \left(1 - \prod_{k=1}^K (1 - (1 - A_{ik}) G_{jk}) \right) \psi_{2,j}(Y_{ij}) \right] \\ &= \sum_{i=1}^N \sum_{j=1}^P \left(\prod_{k=1}^K (1 - (1 - \alpha_{ik}) \gamma_{jk}) \right) \psi_{1,j}(Y_{ij}) + \left(1 - \prod_{k=1}^K (1 - (1 - \alpha_{ik}) \gamma_{jk}) \right) \psi_{2,j}(Y_{ij}) \end{aligned}$$

The second line follows by rewriting the ideal response terms so that the mean-field assumption can be applied to the product.

To ease notational burden we have kept dependence on the slipping and guessing pa-

rameters s and g implicit and will maintain this convention throughout the exposition. As opposed to other parameters, s and g are treated as non-random and estimated via empirical Bayes method.

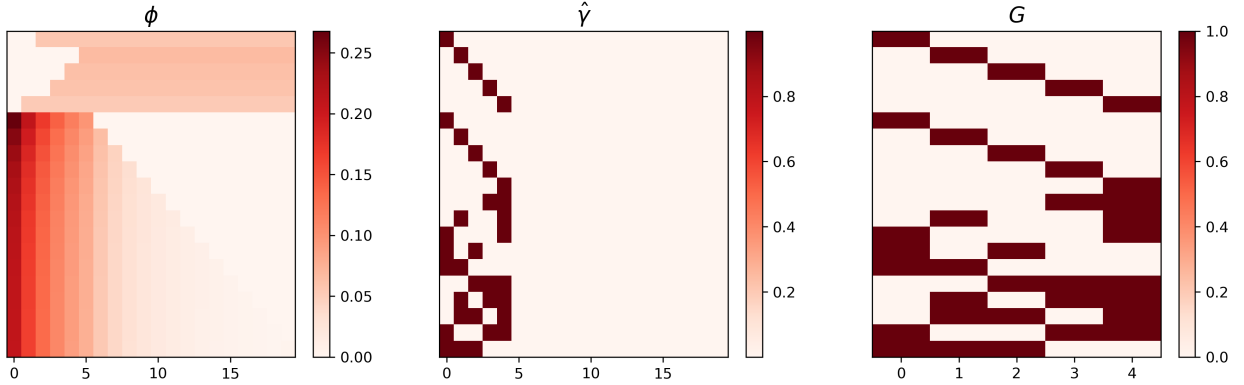


Figure 1: Example run of our algorithm, $s = 0.2$, $g = 0.2$, $N = 1000$, $P = 20$, $K = 5$, $K_{\max} = 20$. *Left*: Estimated ϕ variational parameter, indicating 5 columns are selected. Under the variational parameterization, $\sum_{l=1}^k \phi_{kl}$ is the probability that for skill k , G_{jk} is irrelevant. So, the k th row ϕ_k of ϕ where all of the probability mass is on the $l \leq k$ elements represent skills that are *irrelevant*. For the left panel, the row with $k \geq 6$ represents irrelevant components. *Middle*: Estimated γ variational parameters. *Right*: Ground truth G matrix.

3.1 Cumulative Shrinkage Process to Select the Latent Dimension

The cumulative shrinkage process (CSP; [Legramanti et al., 2020](#)) is a non-parametric prior that determines the number of latent components through data evidence. Variational inference for the CSP prior was previously developed in the context of Gaussian factor models ([Legramanti, 2020](#)). Here, we develop a coordinate ascent variational inference algorithm for the DINA model. We find that in practical problem settings, variational approximations to the CSP are sparse and require minimal hyperparameter tuning. We place a CSP prior on

the matrix $\mathbf{G}$:

$$G_{jk} \sim \text{Bernoulli}\left(\left(1 - \sum_{l=1}^k z_{kl}\right)\frac{1}{2} + \left(\sum_{l=1}^k z_{kl}\right)\delta\right) \quad (5)$$

$$z_k = (z_{k1}, \dots, z_{kK_{\max}}) \sim \text{Categorical}(\omega_1, \dots, \omega_{K_{\max}}), \quad k = 1 : K_{\max} \quad (6)$$

$$\omega_l = \nu_l \prod_{m=1}^{l-1} (1 - \nu_m), \quad l = 1 : K_{\max} \quad (7)$$

$$\nu_l \sim \text{Beta}(1, \kappa), \quad l = 1 : K_{\max} - 1 \quad (8)$$

where ν_l , $l = 1, \dots, K_{\max} - 1$ are stick breaking parameters, z_k is a one-hot vector, and $\nu_{K_{\max}} = 1$. Here, $K_{\max}$ is interpreted as the maximum possible number of latent components, and the true number of latent components is much smaller than $K_{\max}$. There are two user-defined hyperparameters; κ controls the prior expected number of latent components, while δ is a constant close to 0 that controls the probability the item-attribute matrix G has a non-zero value for an inactive component. In our experiments, δ is fixed to $1e - 2$ and κ is fixed to 2, while $K_{\max}$ can be set as the number of features, P . Recovery of the true number latent components is found to be possible across all experimental settings with these fixed choices of hyperparameters.

We now describe the variational approximation with the CSP prior. The latent variables are $\{A_{ik}\}_{i=1, k=1}^{N, K_{\max}}$, $\{G_{jk}\}_{j=1, k=1}^{P, K_{\max}}$, $\{z_k\}_{k=1}^{K_{\max}}$, $\{\nu_k\}_{k=1}^{K_{\max}}$. In addition to the mean field assumption on $\mathbf{A}$ and $\mathbf{G}$ in Eq. (4), we have $\nu_k \sim \text{Beta}(\pi_{0k}, \pi_{1k})$ independently. Finally $z_k \sim \text{Categorical}(\phi_{k1}, \dots, \phi_{kK_{\max}})$. We have $\mathbb{E}_Q[z_k] = \phi_k$, $\sum_l \phi_{kl} = 1$. The joint likelihood is

$$p(\mathbf{Y}, \mathbf{A}, \mathbf{G}, z, \nu; s, g, \delta) = \prod_{i,j} \left\{ p(Y_{ij} | A_i, G_j, s, g) \prod_k \left[p(G_{jk} | z_k, \delta) p(z_k | \nu_{1:K}) p(A_{ik}) \right] \right\} \prod_k p(\nu_k),$$

with latent variables $(\mathbf{A}, \mathbf{G}, z, \nu)$ and model parameters (s, g, δ) . The variational distribution

is

$$Q(\mathbf{A}, \mathbf{G}, z, \nu) = \prod_{i,k} q(A_{ik}; \alpha_{ik}) \prod_{j,k} q(G_{jk}; \gamma_{jk}) \prod_k q(z_k; \phi_k) \prod_k q(\nu_k; \pi_0, \pi_1) \quad (9)$$

Combining this with the expected likelihood term, the evidence lower bound for DINA-CSP is (up to constants):

$$\begin{aligned} \text{ELBO}(\alpha, \gamma, \phi, \pi_0, \pi_1) = & \sum_{i=1}^N \sum_{j=1}^P \left(\prod_{k=1}^{K_{\max}} (1 - (1 - \alpha_{ik})\gamma_{jk}) \right) \psi_{1,j}(Y_{ij}) + \left(1 - \prod_{k=1}^{K_{\max}} (1 - (1 - \alpha_{ik})\gamma_{jk}) \right) \psi_{2,j}(Y_{ij}) \\ & + \sum_{j=1}^P \sum_{k=1}^{K_{\max}} \left((1 - \sum_{l=1}^k \phi_{kl}) \log\left(\frac{1}{2}\right) + \left(\sum_{l=1}^k \phi_{kl}\right) (\gamma_{jk} \log(\delta) + (1 - \gamma_{jk}) \log(1 - \delta)) \right) \\ & + \sum_{k=1}^{K_{\max}} \sum_{l=1}^{K_{\max}} \phi_{kl} \mathbb{E}_{\nu_{1:l}} \log(\nu_l \prod_{m=1}^{l-1} (1 - \nu_m)) + \sum_{k=1}^{K_{\max}} (\kappa - 1) \mathbb{E}_{\nu_k} \log(1 - \nu_k) + H_Q \\ =: & \tilde{L}(\alpha, \gamma, \phi, \pi_0, \pi_1) + H_Q \end{aligned} \quad (10)$$

where expectations are taken under the variational Beta distribution for ν_l ; namely $\nu_l \sim \text{Beta}(\pi_{0,l}, \pi_{1,l})$. The term H_Q denotes the entropy of the distribution Q , i.e., $H_Q := -\int \log Q dQ$, and $\tilde{L}(\alpha, \gamma, \phi, \pi_0, \pi_1)$ includes the non-entropy terms of the ELBO (namely, the expected complete data log-likelihood).

Each group of variational parameters permits a coordinate ascent update in closed form (see Supplementary Material § 2). First-order conditions lead to the following updates for the posterior variational Beta parameters of ν :

$$\pi_{0,k}^{(t)} = 1 + \sum_{l=1}^{K_{\max}} \phi_{l,k}^{(t-1)}, \quad \pi_{1,k}^{(t)} = \kappa + \sum_{l=1}^{K_{\max}} \sum_{m=k+1}^{K_{\max}} \phi_{l,k}^{(t-1)}$$

for $k = 1, \dots, K_{\max} - 1$. For the categorical parameters, first-order conditions lead to the

following update:

$$\phi_{k,l}^{(t)} \propto \begin{cases} \exp \left(\mathbb{E}_{\pi^{(t-1)}} (\log \omega_l) + P \log(1/2) \right), & \text{if } l \geq k+1 \\ \exp \left(\mathbb{E}_{\pi^{(t-1)}} (\log \omega_l) + \sum_{j=1}^P \gamma_{j,k}^{(t-1)} \log(\delta) \right. \\ \quad \left. + (P - \sum_{j=1}^P \gamma_{j,k}^{(t-1)}) \log(1 - \delta) \right), & \text{if } l < k+1 \end{cases} \quad (11)$$

For the item-attribute matrix variational parameters γ , the first order conditions give:

$$\gamma_{jk}^{(t)} = \sigma \left(\sum_{i=1}^N \left(- (1 - \alpha_{ik}^{(t-1)}) \prod_{l \neq k} (1 - (1 - \alpha_{il}^{(t-1)}) \gamma_{jl}^{(t-1)}) \right) \psi_{1,j}(Y_{ij}) \right) \quad (12)$$

$$+ \left((1 - \alpha_{ik}^{(t-1)}) \prod_{l \neq k} (1 - (1 - \alpha_{il}^{(t-1)}) \gamma_{jl}^{(t-1)}) \right) \psi_{2,j}(Y_{ij}) + \left(\sum_{l=1}^k \phi_{kl}^{(t-1)} \log\left(\frac{\delta}{1-\delta}\right) \right) \quad (13)$$

$$= \sigma \left(\nabla_{\gamma_{jk}} \tilde{L}(\alpha^{(t-1)}, \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}) \right).$$

Here, we point out the importance of $\log(\frac{\delta}{1-\delta})$, a quantity of negative and large magnitude, that induces column-wise sparsity. When $\sum_{l=1}^k \phi_{kl}^{(t-1)} = 1$, $\log(\frac{\delta}{1-\delta})$ is added to the likelihood terms for all j , resulting in values of $\gamma_{jk}^{(t)}$ near 0. Hence, it behaves as a penalty that encourages group-wise sparsity.

However, the penalty is adaptive as the parameters $\phi_{k,l}$ are updated. For a given column k of the item-attribute matrix G , if $\gamma_{j,k}^{(t-1)}$ are mostly small for $j \in [P]$, then $(P - \sum_{j=1}^P \gamma_{j,k}^{(t-1)}) \log(1 - \delta)$ in [Eq. \(11\)](#) will be large and positive, so that $\phi_{k,l}^{(t)}$ is larger for $l < k+1$ than for $l \geq k+1$. When $\phi_{k,l}^{(t)}$ is large for values of l that are less than $k+1$, the weight of the penalty $\log(\frac{\delta}{1-\delta})$ in [Eq. \(13\)](#), given by $\sum_{l=1}^k \phi_{kl}^{(t)}$, is large, which encourages group-wise sparsity. Likewise, if $\gamma_{j,k}^{(t-1)}$ are near 1 for all $j \in [P]$, then most of the mass of $\phi_{k,l}^{(t)}$ is on $l \geq k+1$, leading to small values of $\sum_{l=1}^k \phi_{kl}^{(t)}$ and hence a small value of the penalty weight. The CSP variational coordinate ascent updates can be thought of as in similar spirit as adaptive or non-convex penalized methods ([Zhang, 2010](#); [Fan and Li, 2001](#); [Zou, 2006](#); [Ročková and George, 2016](#)).

Finally, updates for α are as $\alpha_{ik}^{(t)} = \sigma\left(\nabla_{\alpha_{ik}} \tilde{L}(\alpha^{(t-1)}, \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)})\right)$, where importantly $\nabla_{\alpha_{ik}} \tilde{L}(\alpha^{(t-1)}, \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)})$ has no dependence on $\alpha_{ik}^{(t-1)}$, which facilitates computation of the update.

Algorithm 1: Coordinate Ascent Variational Inference for DINA-CSP

Data: Input data Y , δ , κ , $K_{\max}$

Result: $\hat{\gamma}$, $\hat{\alpha}$, $\hat{\pi}$, $\hat{\phi}$, $\hat{s}$, $\hat{g}$

1 Initialization via factor analysis;

2 **while** *not converged* **do**

3 $\gamma_{jk}^{(t)} = \sigma\left(\nabla_{\gamma_{jk}} \tilde{L}(\alpha, \gamma, \phi, \pi_0, \pi_1)\right)$;

4 $\alpha_{ik}^{(t)} = \sigma\left(\nabla_{\alpha_{ik}} \tilde{L}(\alpha, \gamma, \phi, \pi_0, \pi_1)\right)$;

5 $\phi_{k,l}^{(t)} \propto \exp\left(\mathbb{E}_{\pi^{(t-1)}}(\log \omega_l) + p \log(1/2)\right)$ for $l \geq k+1$;

6 $\phi_{k,l}^{(t)} \propto$
 $\exp\left(\mathbb{E}_{\pi^{(t-1)}}(\log \omega_l) + \sum_{j=1}^P \gamma_{j,k}^{(t-1)} \log(\delta) + (P - \sum_{j=1}^P \gamma_{j,k}^{(t-1)}) \log(1-\delta)\right)$ for $l < k+1$
 ;

7 normalize $\phi_k^{(t)} \leftarrow \phi_k^{(t)} / \sum_{l=1}^{K_{\max}} \phi_{kl}^{(t)}$;

8 $\pi_{0,k}^{(t)} = 1 + \sum_{l=1}^{K_{\max}} \phi_{l,k}^{(t-1)}$;

9 $\pi_{1,k}^{(t)} = \kappa + \sum_{l=1}^{K_{\max}} \sum_{m=k+1}^{K_{\max}} \phi_{l,m}^{(t-1)}$;

10 Updates for s and g ;

11 $s_j^{(t)} = \frac{\sum_{i=1}^N (1-Y_{ij}) \left(\prod_{k=1}^{K_{\max}} (1 - (1-\alpha_{ik}^{(t-1)}) \gamma_{jk}^{(t-1)}) \right)}{\sum_{i=1}^N \prod_{k=1}^{K_{\max}} (1 - (1-\alpha_{ik}^{(t-1)}) \gamma_{jk}^{(t-1)})}$;

$g_j^{(t)} = \frac{\sum_{i=1}^N Y_{ij} \left(1 - \prod_{k=1}^{K_{\max}} (1 - (1-\alpha_{ik}^{(t-1)}) \gamma_{jk}^{(t-1)}) \right)}{\sum_{i=1}^N \left(1 - \prod_{k=1}^{K_{\max}} (1 - (1-\alpha_{ik}^{(t-1)}) \gamma_{jk}^{(t-1)}) \right)}$;

12 **return** $\hat{\gamma}$, $\hat{\alpha}$, $\hat{\pi}$, $\hat{\phi}$, $\hat{s}$, $\hat{g}$;

3.2 Full Parameterization of the Attribute Mastery Matrix

In some cases, the mean-field assumption for the Attribute Mastery Matrix $\mathbf{A}$ may be too restrictive. Instead, we can use a variational posterior for A_i that encodes possible dependencies among latent components. We introduce an auxiliary integer-valued variable, $\tau_i \in [0, \dots, 2^{K_{\max}} - 1]$, taking on values according to variational distribution;

$$P(\tau_i = b) = \chi_{ib}; \quad b = 0, \dots, 2^{K_{\max}} - 1,$$

Denote $\text{bin}_K : [0, \dots, 2^K - 1] \rightarrow \{0, 1\}^K$ as the binary representation function for an integer where $\text{bin}_K(n)_i = \lfloor \frac{n}{2^{K-i-1}} \rfloor \bmod 2$. Then $A_i = \text{bin}_{K_{\max}}(\tau_i)$, hence requiring $N2^{K_{\max}}$ latent parameters instead of $NK_{\max}$. Such a parameterization may be suitable when $K_{\max}$ is relatively small. With this parameterization, full coordinate ascent updates are available. Of note, the ELBO under this parameterization can be written (up to constants)

$$\begin{aligned}
& \text{ELBO}_{\text{full}}(\chi, \gamma, \phi, \pi_0, \pi_1) \\
&= \sum_{i=1}^N \sum_{b=1}^{2^{K_{\max}}} \left\{ \chi_{ib} \sum_{j=1}^P \left(\prod_{k=1}^{K_{\max}} (1 - (1 - \text{bin}_{K_{\max}}(b)_k) \gamma_{jk}) \right) \psi_{1,j}(Y_{ij}) \right. \\
&+ \left. \left(1 - \prod_{k=1}^{K_{\max}} (1 - (1 - \text{bin}_{K_{\max}}(b)_k) \gamma_{jk}) \right) \psi_{2,j}(Y_{ij}) \right\} \\
&+ \sum_{j=1}^P \sum_{k=1}^{K_{\max}} \left(\left(1 - \sum_{l=1}^k \phi_{kl} \right) \log\left(\frac{1}{2}\right) + \left(\sum_{l=1}^k \phi_{kl} \right) (\gamma_{jk} \log(\delta) + (1 - \gamma_{jk}) \log(1 - \delta)) \right) \\
&+ \sum_{k=1}^{K_{\max}} \sum_{l=1}^{K_{\max}} \phi_{kl} \mathbb{E}_Q \log(\nu_l \prod_{m=1}^{l-1} (1 - \nu_m)) + \sum_{k=1}^{K_{\max}} (\kappa - 1) \mathbb{E}_Q \log(1 - \nu_k) + H_Q \\
&= \tilde{L}_{\text{full}}(\chi, \gamma, \phi, \pi_0, \pi_1) + H_Q
\end{aligned} \tag{14}$$

This expression is an average of the original ELBO over the distribution parameterized by χ . Simple calculations show that coordinate ascent updates for ϕ , π_0 , and π_1 remain the same as in Algorithm 1.

For γ , first-order conditions yield the analogous update

$$\gamma_{jk}^{(t)} \leftarrow \sigma \left(\nabla_{\gamma_{jk}} \tilde{L}_{\text{full}}(\chi^{(t-1)}, \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}) \right).$$

However, this update requires a sum over $2^{K_{\max}}$ components. To improve computational efficiency, we approximate the summation over $2^{K_{\max}}$ terms in Eq. (14) with the summand corresponding to the largest value of χ_{ib} over b . This approximation is equivalent to approximating $\tilde{L}_{\text{full}}(\chi, \gamma, \phi, \pi_0, \pi_1)$ with the log-likelihood $\tilde{L}(\alpha, \gamma, \phi, \pi_0, \pi_1)$ of Eq. (10) with α_i

equals to the one-hot encoded $\arg \max_b \chi_{ib}$.

The update for χ_i is of the form:

$$\chi_i^{(t)} = \text{softmax} \left(\tilde{L}_i(\text{bin}_{K_{\max}}(0), \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}), \dots, \right. \\ \left. \tilde{L}_i(\text{bin}_{K_{\max}}(2^{K_{\max}} - 1), \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}) \right)$$

where $\tilde{L}_i$ is the subset of terms in the expression of $\tilde{L}$ that depends on α_i . Hence, updating χ_i proceeds by testing every possible binary pattern for $\mathbf{A}_i$ and constructing probability weights based on the ELBO. Empirically, we find that the full variational parameterization of the matrix $\mathbf{A}$ improves the recovery of latent parameters (§ 6.2).

Algorithm 2: Full Coordinate Ascent Variational Inference for DINA-CSP

Data: Input data Y , δ , κ , $K_{\max}$

Result: $\hat{\gamma}$, $\hat{\chi}$, $\hat{\pi}$, $\hat{\phi}$, $\hat{s}$, $\hat{g}$

1 Initialization via factor analysis;

2 **while** *not converged* **do**

3 $\chi_i^{(t)} = \text{softmax} \left(\tilde{L}_i(\text{bin}_{K_{\max}}(0), \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}), \dots, \tilde{L}_i(\text{bin}_{K_{\max}}(2^{K_{\max}} - 1), \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}) \right) ;$

4 $m_i \leftarrow \arg\max(\chi_i^{(t)}) ;$

5 $\gamma_{jk}^{(t)} = \sigma \left(\nabla_{\gamma_{jk}} \tilde{L}(\text{bin}_{K_{\max}}(m), \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}) \right) ;$

6 Update π , s , g , ϕ as in Algorithm 1.

7 **return** $\hat{\gamma}$, $\hat{\chi}$, $\hat{\pi}$, $\hat{\phi}$, $\hat{s}$, $\hat{g}$;

3.3 Updates for Correlated Discrete Latent Variables

Realistic settings have correlated latent components, i.e. $\text{corr}(A_{ik}, A_{il}) > 0$ for $k \neq l$. This can be encoded by the following hierarchical prior for A_i :

$$p_i \sim \text{Beta}(\rho, \rho), \quad A_{ik} | p_i \sim \text{Bernoulli}(p_i)$$

Under this model, marginally $A_{ik} \sim \text{Bernoulli}(1/2)$, as in the original model; however, $\text{corr}(A_{ik}, A_{il}) = \frac{1}{2\rho+1}$ for $k \neq l$. In the limit $\rho \rightarrow \infty$, the prior converges to a point mass at $1/2$ and we recover the original model. As $\rho \rightarrow 0$, p_i is concentrated near 0 and 1 and the latent components become correlated. Empirically, we find that the use of a correlated prior for $\mathbf{A}$ improves recovery of latent parameters (§ 6.2).

Assuming a mean-field posterior for p_i : $p_i \sim \text{Beta}(q_{i,a}, q_{i,b})$, the following updates can be included in Algorithm 1:

$$\begin{aligned} q_{i,a}^{(t)} &= \rho + \sum_{k=1}^{K_{\max}} \alpha_{ik}^{(t-1)} \\ q_{i,b}^{(t)} &= \rho + K_{\max} - \sum_{k=1}^{K_{\max}} \alpha_{ik}^{(t-1)} \\ \alpha_{ik}^{(t)} &= \sigma \left(\nabla_{\alpha_{ik}} \tilde{L}(\alpha^{(t-1)}, \gamma^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}) + \Psi(q_{i,a}^{(t-1)}) - \Psi(q_{i,b}^{(t-1)}) \right) \end{aligned}$$

where $\Psi(x)$ is the digamma function. Given an estimate of $\mathbf{A}$, $\hat{\mathbf{A}}$, and K , $\hat{K}$, e.g. from Algorithm 1, an empirical Bayes estimate of ρ can be determined via:

$$\hat{\rho} = \frac{1}{2} \left(\frac{1}{\widehat{\text{corr}}} - 1 \right)_+$$

where $\widehat{\text{corr}} := \frac{2}{\hat{K}(\hat{K}-1)} \sum_{k=1}^{\hat{K}} \sum_{l=k+1}^{\hat{K}} \text{corr}(\hat{A}_{ik}, \hat{A}_{il})$.

4 Practical Considerations of the Algorithms

We now discuss how to set the initialization and compute the gradient updates to improve the robustness of the variational algorithm.

4.1 Spectral Initialization with Varimax Rotation

The varimax rotation was originally developed by Kaiser (Kaiser, 1958) and has been used ubiquitously in many applied contexts (Abdi, 2003; Shrestha, 2021). Recent work Rohe and Zeng (2023) has shown that principal components analysis with appropriate kurtosis maximizing rotation performs statistical inference under a semiparametric factor model. First we point out that in the special case where each item requires only one latent skill, the DINA model coincides with a factor model. Here, varimax rotations of principal components should estimate the item-attribute matrix well. Under identifiability assumptions (Gu and Xu, 2019, 2021), such items exist in the data.

We initialize by applying a classical SVD based factor analysis estimate of $\mathbf{G}$ (Barber (2012), *Algorithm 21.1*), appropriately rotated with the varimax rotation. This yields an initialization for γ . We then initialize α by applying the update for α_{ik} in Algorithm 1 until convergence with γ fixed (Supplementary Material § 3). Spectral initialization of methods susceptible to local optima has been studied in Zhang et al. (2014). Our initialization is motivated by approximation of the DINA ideal response with a linear factor analysis model. By binomial expansion of the ideal response term:

$$\begin{aligned} \prod_{k=1}^K (1 - (1 - A_{ik})G_{jk}) &= \sum_{t=0}^K \sum_{S \in \binom{[K]}{t}} (-1)^t \prod_{k \in S} (1 - A_{ik})G_{jk} \\ &= 1 - \sum_{k=1}^K (1 - A_{ik})G_{jk} + \sum_{t=2}^K \sum_{S \in \binom{[K]}{t}} (-1)^t \prod_{k \in S} (1 - A_{ik})G_{jk} \end{aligned}$$

Letting $\mathbf{J} \in \{0, 1\}^{N \times P}$ be the ideal response term with $J_{ij} = \prod_{k=1}^K (1 - (1 - A_{ik})G_{jk})$, we see that the rank of $\mathbf{J} - \mathbf{1}\mathbf{1}^\top$ is less than or equal to the number of unique subsets $S \subset [K]$, with $|S| > 0$ and $\prod_{k \in S} G_{jk} = 1$ for some j ; i.e. the number of unique rows of the item-attribute matrix. In cases where such S all have $|S| = 1$, then the ideal response reduces to a linear

factor model:

$$J_{ij} = 1 - \sum_{k=1}^K (1 - A_{ik}) G_{jk}, \quad \mathbf{1}\mathbf{1}^\top - \mathbf{J} = (\mathbf{1}\mathbf{1}^\top - \mathbf{A})\mathbf{G}^\top.$$

Empirically, we find that this initialization leads to vast improvements in optimization performance. We show that an optimal linear approximation to the non-linear DINA model has the correct sparsity pattern in the special case where each A_{ik} is independently generated from a Bernoulli($\frac{1}{2}$) distribution. This offers some intuitive insight into our empirical observation that initialization via linear factor model improves the convergence of our DINA estimation algorithm.

4.2 Stochastic Gradient Estimation with Dropout

A challenge for mean-field variational inference in optimizing non-convex evidence lower bounds is the presence of saddle points or local optima. Motivated by the capability of stochastic gradients in optimizing non-convex objectives, the stability of DINA-CSP objective can be further improved by following noisy gradient estimates rather than the exact coordinate ascent. We introduce randomness to the gradient estimation by dropout ([Srivastava et al., 2014](#)) by first adding noise to α and γ to form $\tilde{\alpha}$ and $\tilde{\gamma}$:

$$\tilde{\alpha}_{ik} = \begin{cases} \alpha_{ik} & \text{w.p. } 1 - \zeta; \\ U_{ik} & \text{w.p. } \zeta. \end{cases} \quad \tilde{\gamma}_{jk} = \begin{cases} \gamma_{jk} & \text{w.p. } 1 - \zeta; \\ U'_{jk} & \text{w.p. } \zeta, \end{cases}$$

where $U_{ik}, U'_{jk} \sim \text{Unif}[0, 1]$. Then, updates for α and γ are formed based on $\tilde{\alpha}$ and $\tilde{\gamma}$:

$$\begin{aligned} \gamma_{jk}^{(t)} &= \sigma \left(\nabla_{\gamma_{jk}} \tilde{L}(\tilde{\alpha}^{(t-1)}, \tilde{\gamma}^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}) \right) \\ \alpha_{ik}^{(t)} &= \sigma \left(\nabla_{\alpha_{ik}} \tilde{L}(\tilde{\alpha}^{(t-1)}, \tilde{\gamma}^{(t-1)}, \phi^{(t-1)}, \pi_0^{(t-1)}, \pi_1^{(t-1)}) \right) \end{aligned}$$

Fig. 3 and Supplementary Fig. 1 show that these estimates improve convergence.

5 Theoretical Results

We develop three theoretical results regarding variational inference for the DINA model. First, we show that the sparsity pattern of the true item-attribute matrix $\mathbf{G}$ is preserved by the best linear approximation to the item-attribute matrix. Second, we establish the sample complexity of obtaining the true values of $\mathbf{A}$ and $\mathbf{G}$ as fixed points of the variational inference coordinate ascent algorithm. Third, we establish the minimax error under these settings, showing rate optimality up to log factors of our approach.

Proposition 1 (Linear approximation to ideal response). *Let $A_{ik} \sim_{iid} \text{Bernoulli}(1/2)$, and let γ_N be defined as:*

$$\gamma_N = \underset{\gamma}{\operatorname{argmin}} \frac{1}{NP} \sum_{i=1}^N \sum_{j=1}^P \left(\prod_{k=1}^K (1 - (1 - A_{ik})G_{jk}) - \sum_{k=1}^K A_{ik}\gamma_{jk} \right)^2$$

Then $\gamma_N \rightarrow \gamma^$ as $N \rightarrow \infty$ almost surely, with $\mathbf{1}\{\gamma_{jk}^* > 0\} = G_{jk}$ for all $j = 1, \dots, P$ and $k = 1, \dots, K$. For j such that $\sum_{k=1}^K G_{jk} = 1$, we have $\gamma_{jk}^* = G_{jk}$ for all $k \in [K]$.*

Proposition 1 asserts that the sparsity pattern of an approximation, γ_N , of $\mathbf{G}$ is equal to $\mathbf{G}$ itself. Here, γ_N is chosen to approximate the ideal response $\prod_{k=1}^K (1 - (1 - A_{ik})G_{jk})$ with the low rank matrix with entries given by $\sum_{k=1}^K A_{ik}\gamma_{N,jk}$. This approximation is under the idealized setting where each A_{ik} is $\text{Bernoulli}(1/2)$. This result suggests that initialization via low rank factorization methods may reasonably approximate the true item-attribute matrix and allow for avoidance of pathological fixed points in the variational inference objective function.

The rest of theory results focus on estimation in with general $\mathbf{A}$ and $\mathbf{G}$ under the noisy setting and with slipping and guessing parameters s and g . We begin by establishing identifiability assumptions for $\mathbf{A}$ and $\mathbf{G}$. In order to estimate A_{ik} , which represents whether student

i has latent skill k , we must have a sufficient number of questions in the dataset for which the student has all skills necessary to answer the question *besides* skill k . These questions are the ones that can indicate whether or not the student has skill k . Specifically, we require that there are $\Omega(P)$ such questions, which is formalized in [Assumption 1](#). Analogously, in order to test whether a question requires a certain skill k , we need to have a sufficient number of students who have all skills required to answer the question *in addition to* the skill k . Such students must also not have skill k , so that their answers test whether the question requires skill k . We require $\Omega(N)$ such students per question, which is formalized in [Assumption 2](#).

Assumption 1. For all $(i, k) \in [N] \times [K]$

$$\sum_{j=1}^P G_{jk} \prod_{l \neq k} A_{il}^{G_{jl}} = \Omega(P)$$

This states that for any individual and latent skill there are $\Omega(P)$ number of questions in the questionnaire that the individual can “almost answer” in the sense that they possess all latent skills besides the skill in question. Note that this is automatically satisfied when there are $\Omega(P)$ questions per skill that are unique to that skill. Further, for sufficiently large P let:

$$\frac{1}{P} \sum_{j=1}^P G_{jk} \prod_{l \neq k} A_{il}^{G_{jl}} \geq \Xi$$

for some positive constant $\Xi \in (0, 1)$.

Assumption 2. For all $(j, k) \in [P] \times [K]$

$$\sum_{i=1}^N (1 - A_{ik}) \prod_{l \neq k} A_{il}^{G_{jl}} = \Omega(N)$$

Further, for sufficiently large N let :

$$\frac{1}{N} \sum_{i=1}^N (1 - A_{ik}) \prod_{l \neq k} A_{il}^{G_{jl}} \geq \Xi$$

for some positive Ξ . This is the analogous assumption to Assumption 1, i.e. assuming adequate sampling for every item and latent across individuals.

Example 1. Assumptions 1 and 2 are easy to satisfy and we give some concrete examples...

If G contains blocks of $(1, 1, 0; 1, 0, 1; 0, 1, 1)$ and $A_i = (1, 1, 0)$, then for $j = 1$ and $k = 3$.

For $G_{j,k} = 0$,

We note that $\prod_{l \neq k} (1 - A_{il}^{G_{jl}})$, the leave-one-out ideal response, measures whether student i has all required skills to answer question j besides the k 'th skill. Ξ can be thought of as the minimum (across i, k) fraction of questions (resp. students) that can almost be answered by student i if not for skill k (and hence is a number less than 1). As a simple example of when Assumption 1 is satisfied, consider the case where $G = [I; \dots; I]^\top$ for m copies of the identity matrix $I \in \mathbb{R}^{K \times K}$. Here the number of questions P is equal to mK . Since $\prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = 1$ whenever $G_{jk} = 1$ for all (i, j, k) , we see immediately that Assumption 1 holds with $\Xi = \frac{1}{K}$. Finally, we note that Assumption 1 and Assumption 2 can be thought of as identifiability assumptions. Suppose $\sum_{j=1}^P G_{jk} \prod_{l \neq k} A_{il}^{G_{jl}} = 0$. That is, there are no questions that test whether student i has skill k . Let $\tilde{A}_{i'k'} = A_{i'k'}$ for all $(i', k') \neq (i, k)$ and $\tilde{A}_{ik} = 1$, $A_{ik} = 0$. Further let $\tilde{\mathbf{A}}$ be the matrix with entry (i, k) equal to $\tilde{A}_{ik}$. Then: $\mathbb{P}(\mathbf{Y}; \mathbf{A}, \mathbf{G}) = \mathbb{P}(\mathbf{Y}; \tilde{\mathbf{A}}, \mathbf{G})$. Hence the data distribution under $A_{ik} = 1$ is indistinguishable from the data distribution $A_{ik} = 0$.

Under Assumption 1 and Assumption 2, we establish the sample complexity of the true parameters emerging as fixed points of the DINA-CSP algorithm. We make additional simplifying assumptions that facilitate exposition of our results. $\mathbf{A}$ and $\mathbf{G}$ are the main

parameters of interest, while s_j and g_j are noise parameters that determine the problem difficulty. We assume that s_j and g_j are known and Algorithm 1 is run using the ground truth values of s_j and g_j . Further we assume a constant slipping and guessing rate; that is, $s_j = g_j =: s$ and further $s < \frac{1}{2}$. Finally we assume $K_{\max} = K$, and do not distinguish between K and $K_{\max}$ in this section. This assumption is required simply because A_{ik} is entirely non-identifiable when $G_{jk} = 0$ for all $j \in [P]$; however, the theory can potentially be adapted to the setting where $K_{\max} > K$. With data $\mathbf{Y}$ generated under parameters $\mathbf{A}$ and $\mathbf{G}$, [Theorem 1](#) answers the question of how many data points are necessary to have $\mathbf{A}$ and $\mathbf{G}$ as the fixed points of DINA-CSP with high probability. [Theorem 2](#) establishes the minimax lower bound of estimation of $\mathbf{A}$ and $\mathbf{G}$ under the same assumptions as [Theorem 1](#). Our proof techniques for [Theorem 1](#) resemble those of [Zhang et al. \(2014\)](#) while our proof for [Theorem 2](#) is an application of Le Cam's method.

Theorem 1 (Sample complexity of true parameters being fixed points). *We first define some fundamental constants that determine the difficulty of the problem setting. For simplicity of notation, we let $s_j = g_j = s$ for all $j = 1, \dots, P$ and introduce the assumption $\epsilon := \frac{1}{2} - s > 0$. ϵ quantifies one aspect of the problem difficulty, namely the level of noise in the problem. Additionally, let $\rho := \log(\frac{1-s}{g}) > 0$, which is a function of ϵ under the assumptions. Let $\mathcal{C}(\delta) = \frac{1-\delta}{\delta}$. Further, let Ξ be the constant defined in Assumptions 1 and 2. Under Assumptions 1 and 2, suppose for all $(i, j, k) \in [N] \times [P] \times [K]$:*

$$|\alpha_{ik} - A_{ik}| \leq 1 - (1 - \frac{\epsilon\Xi}{2})^{1/K}, \quad |\gamma_{jk} - G_{jk}| \leq 1 - (1 - \frac{\epsilon\Xi}{2})^{1/K}$$

and the number of items and respondents satisfy:

$$\begin{aligned} N &\geq \frac{1}{\epsilon\Xi} \left(\frac{2}{\rho} \log \left(\frac{\mathcal{C}(\delta)}{1 - (1 - \epsilon\Xi/2)^{1/K}} \right) + \frac{2}{\rho} \log \left(\frac{\mathcal{C}(\delta)}{\mathcal{M}} \right) + \left(\frac{2}{3} + \frac{2}{\epsilon} \right) \log \left(\frac{2PK}{\xi} \right) \right) \\ P &\geq \frac{1}{\epsilon\Xi} \left(\frac{2}{\rho} \log \left(\frac{1}{1 - (1 - \epsilon\Xi/2)^{1/K}} \right) + \frac{2}{\rho} \log \left(\frac{1}{\mathcal{M}} \right) + \left(\frac{2}{3} + \frac{2}{\epsilon} \right) \log \left(\frac{2NK}{\xi} \right) \right) \end{aligned}$$

Then for estimates obtained via a finite number of iterations of CAVI-CSP, with probability at least $1 - \xi$, for all $(i, j, k) \in [N] \times [P] \times [K]$:

$$|\gamma_{jk} - G_{jk}| < \mathcal{M}, \quad |\alpha_{ik} - A_{ik}| < \mathcal{M}$$

for arbitrarily small $\mathcal{M}$. In summary, we conclude that up to log factors, the number of samples necessary is:

$$N = \tilde{\Omega}\left(\frac{1}{\epsilon\Xi \min(\epsilon, \rho)}\right), \quad P = \tilde{\Omega}\left(\frac{1}{\epsilon\Xi \min(\epsilon, \rho)}\right).$$

Next, we establish the minimax lower bound of the estimation problem, which shows that recovery in $o(\frac{1}{\epsilon\rho\Xi})$ samples is not possible.

Theorem 2 (Minimax lower bound). *We show that $\epsilon\rho\Xi$ is the fundamental quantity that determines the sample complexity. Let the parameter space Θ be the set of $\mathbf{A}$ and $\mathbf{G}$ such that Assumptions 1 and 2 hold (with constant Ξ). Then:*

- If $P \leq \kappa \frac{1}{\epsilon\rho\Xi}$ for $0 < \kappa < 1$, then:

$$\inf_{\hat{\mathbf{A}}} \sup_{\mathbf{A}, \mathbf{G} \in \Theta} \mathbb{P}(\hat{\mathbf{A}} \neq \mathbf{A}) \geq \frac{1}{4}(1 - \sqrt{\kappa})$$

- If $N \leq \kappa \frac{1}{\epsilon\rho\Xi}$ for $0 < \kappa < 1$, then:

$$\inf_{\hat{\mathbf{G}}} \sup_{\mathbf{A}, \mathbf{G} \in \Theta} \mathbb{P}(\hat{\mathbf{G}} \neq \mathbf{G}) \geq \frac{1}{4}(1 - \sqrt{\kappa})$$

Combining Theorems 1 and 2, we establish that the variational inference algorithm achieves the minimax rate up to a factor of $\frac{\min(\rho, \epsilon)}{\epsilon}$ and log terms. For small ϵ , we have $\lim_{\epsilon \rightarrow 0} \frac{\rho}{\epsilon} = 4$ and hence $\rho = \Theta(\epsilon)$ and the rates in Theorems 1 and 2 match. There are a few caveats to our theoretical results. These caveats are directions of possible further inquiry. First, we deal with the sample complexity of uncovering the true parameters as fixed points of our algorithm rather than guaranteed recovery. In general, because of the non-convexity of the DINA-CSP objective, the algorithm is not guaranteed to recover the true parameters $\mathbf{A}$ and $\mathbf{G}$. However, one could determine whether spectral initialization combined with CAVI achieves guaranteed recovery at the rate in [Theorem 2](#) (as in [Zhang et al. \(2014\)](#)). Second, to simplify the results we assume that slipping and guessing parameters are known and used in the algorithm updates. This restriction can be removed, as accurate estimation of s_j and g_j is possible as long as initialization of (α, γ) is close to $(\mathbf{A}, \mathbf{G})$. Finally, our theory does not address the groupwise variable selection problem where $K_{\max} > K$.

6 Simulation Studies

6.1 Simulations for an unknown number of latent components

We examine the performance of DINA-CSP for various sample size $N \in \{500, 1000, 2000\}$, the true underlying latent dimension $K \in \{3, 4, 5, 10\}$, and the correlation between the attributes $\rho \in \{0, 0.25, 0.5\}$. The slip and guessing parameters are set to 0.2 in all situations.

The item-attribute matrix $\mathbf{G}$ is specified as Table 1 in [Chen et al. \(2021\)](#) for $K \leq 5$ which corresponds to the size $(K, P) = (3, 18), (4, 18), (5, 20)$, respectively. For $K > 5$, the item-attribute matrix design follows the simulation study of [Gu and Xu \(2023\)](#) with $(K, P) = (10, 30)$. For each simulation setting, the true $\mathbf{G}$ matrix contains $P/(2K)$ copies of $\mathbf{I}_K$ as sub-matrices stacking vertically, and $K_{\max}$ is set as P .

For the latent attribute profile matrix $\mathbf{A}$, we first generate a vector $\boldsymbol{\theta}_i \sim \mathcal{N}(0, \boldsymbol{\Sigma}_K)$ where

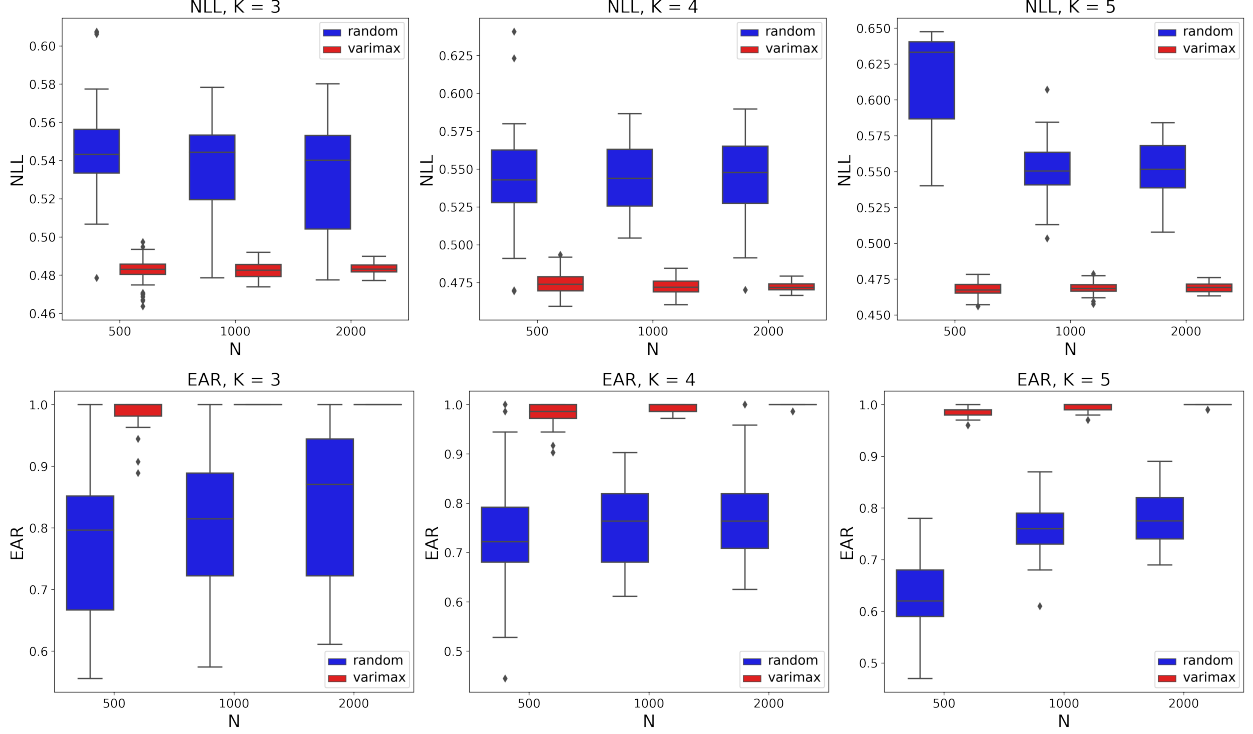


Figure 2: *Initialization improves algorithm performance.* DINA negative log likelihood (top row) and Elementwise accuracy rate (bottom row) across a number of experimental settings evaluated at the VI posterior mode for both uniform random initialization and initialization via the factor analysis with varimax. Here, we evaluate the base variational parameterization, with $\rho = 0, s = 0.2, g = 0.2$.

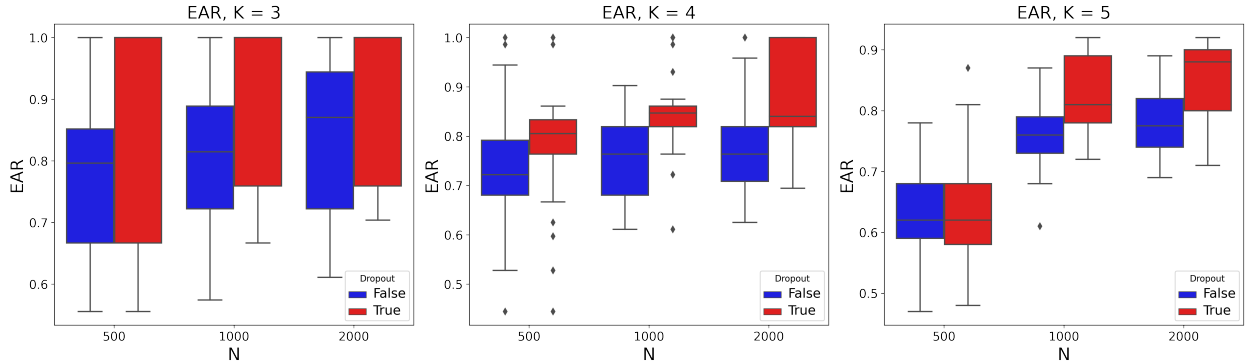


Figure 3: *Dropout improves algorithm performance with random initialization.* Elementwise accuracy rate (EAR) across a number of experimental settings and models evaluated at the VI posterior mode with dropout and without dropout. Here, we evaluate the base variational parameterization with uniform random initialization, with $\rho = 0, s = 0.2, g = 0.2$.

$\Sigma_K = (1 - \rho)\mathbf{I}_K + \rho(\mathbf{1}_K\mathbf{1}_K^\top)$ is the covariance matrix with the diagonal elements as 1 and all the other elements as ρ (Chen et al., 2018). Following Chen et al. (2021), each latent

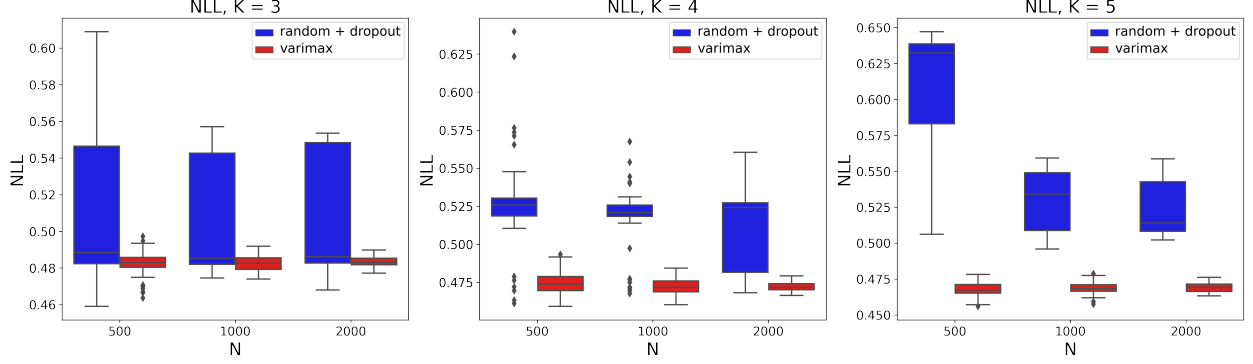


Figure 4: *Varimax initialization without dropout is superior to uniform random initialization with dropout.* Negative log likelihood (NLL) across a number of experimental settings evaluated at the VI posterior mode with varimax initialization or with random uniform initialization with dropout. Here, we evaluate the base variational parameterization with $\rho = 0, s = 0.2, g = 0.2$.

attribute α_i is generated by

$$\alpha_{ik} = \begin{cases} 1 & \text{if } \theta_{ik} > \Phi^{-1}(\frac{k}{K+1}) \\ 0 & \text{otherwise,} \end{cases} \quad (15)$$

where Φ^{-1} is the inverse cumulative distribution function of a standard normal distribution. This construction encodes a heterogeneous difficulty of the latent attributes where the attribute with a larger index k is more difficult to master.

The average recovery rates are presented in Table 1 (Fig. 1). We compute the average accuracy rate to estimate the number of attributes K . We evaluate the elementwise accuracy rate (EAR),

$$\text{EAR} = \frac{1}{PK} \sum_{j=1}^P \sum_{k=1}^K \mathbb{1}[\hat{G}_{jk} = G_{jk}]. \quad (16)$$

The EAR is defined as the proportion of the original true $\mathbf{G}$ matrix that is estimated correctly given the closest K columns of $\hat{\mathbf{G}}$. We also compute the number of over-specified elements

(NOSE) to measure the false discovery (Chen et al., 2021)

$$\text{NOSE} = \sum_{k=K+1}^{\hat{K}} \hat{G}_{jk}. \quad (17)$$

All the results are averaged over 100 independent trials. The results for a large number of latent attributes K are presented in Table 2 (Fig. 5). We consider the conditions $(K, P) \in \{(15, 50), (25, 80), (35, 100)\}$ and $\rho \in \{0, 0.25\}$. For all conditions, we set $N = 2000$ and $K_{\max} = 50$. Table 3 compares the estimation accuracy of the latent attributes K between CAVI-CSP and the Crimp sampler (Chen et al., 2021).

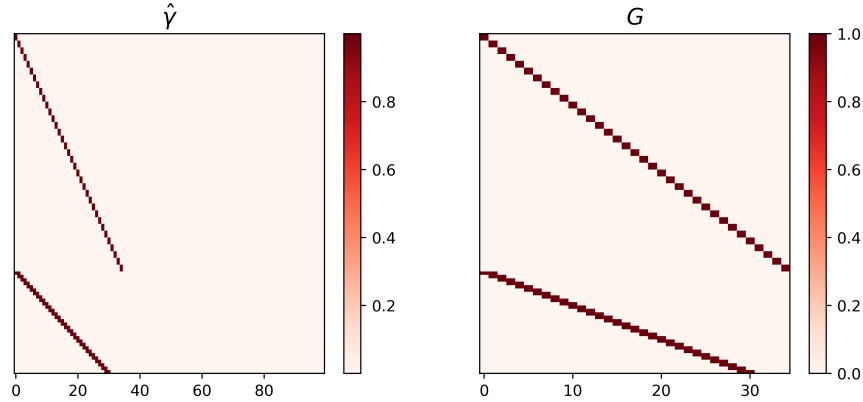


Figure 5: Example simulation run of variational inference procedure on large scale data, demonstrating perfect recovery in a setting with typical levels of noise. for $s = 0.2$, $g = 0.2$, $N = 3000$, $P = 100$, $K = 35$, $K_{\max} = 100$. *Left*: Estimated γ variational parameters. *Right*: Ground truth $\mathbf{G}$ matrix.

Furthermore, the variational inference strategy is substantially faster than the Crimp sampler, with all experiments running in less than 20 seconds on average (Fig. 6, left). Due to the mean-field parameterization, the dependence of run time on K does not scale quickly (Fig. 6, right).

K	N	$\hat{K}_{est.}$	EAR	NOSE	K	N	$\hat{K}_{est.}$	EAR	NOSE
$\rho = 0$									
3	500	95%	99.30%	0.09	5	500	100%	98.50%	0
	1000	99%	100.00%	0.02		1000	100%	99.50%	0
	2000	100%	100.00%	0		2000	100%	99.90%	0
4	500	100%	98.20%	0	10	500	100%	99.40%	0
	1000	100%	99.50%	0		1000	100%	99.70%	0
	2000	100%	100.00%	0		2000	100%	100.00%	0
K	N	$\hat{K}_{est.}$	EAR	NOSE	K	N	$\hat{K}_{est.}$	EAR	NOSE
$\rho = 0.25$									
3	500	99%	86.30%	0.06	5	500	88%	88.80%	0.18
	1000	100%	86.80%	0		1000	99%	89.60%	0
	2000	100%	86.80%	0		2000	100%	89.40%	0
4	500	95%	86.10%	0.091	10	500	39%	92.50%	5
	1000	99%	86.20%	0		1000	67%	95.10%	1.16
	2000	100%	85.80%	0		2000	73%	96.10%	0.07

Table 1: Average recovery rate under each condition.

K	$\hat{K}_{est.}$	EAR	NOSE	K	$\hat{K}_{est.}$	EAR	NOSE
$\rho = 0$				$\rho = 0.25$			
15	100%	100.00%	0	15	12%	94.30%	35.48
25	92%	100.00%	0.08	25	24%	97.50%	25.36
35	100%	100.00%	0	35	0%	98.80%	6.333

Table 2: Results for the setting with a large number of latent attributes K .

K	N	$\hat{K}_{est.}$ CSP	$\hat{K}_{est.}$ Crimp	K	N	$\hat{K}_{est.}$ CSP	$\hat{K}_{est.}$ Crimp	K	N	$\hat{K}_{est.}$ CSP	$\hat{K}_{est.}$ Crimp
$\rho = 0$				$\rho = 0.25$				$\rho = 0.5$			
3	500	96%	96%	3	500	99%	82%	3	500	100%	79%
	1000	99%	92%		1000	100%	85%		1000	100%	74%
	2000	100%	90%		2000	100%	87%		2000	100%	74%
4	500	100%	99%	4	500	95%	80%	4	500	84%	76%
	1000	100%	90%		1000	99%	68%		1000	87%	65%
	2000	100%	76%		2000	100%	65%		2000	95%	37%
5	500	100%	90%	5	500	88%	41%	5	500	62%	37%
	1000	100%	65%		1000	99%	32%		1000	75%	20%
	2000	100%	57%		2000	100%	21%		2000	79%	15%

Table 3: Comparison of K estimate.

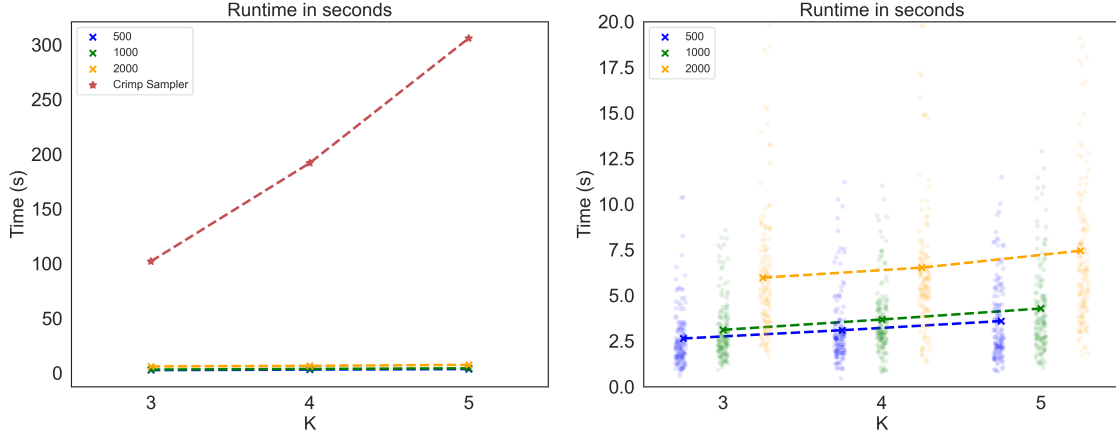


Figure 6: Run time of CAVI algorithm relative to Crimp sampler for experiments in table. *Left:* Run times for CAVI are substantially shorter than the Crimp sampler across the same experimental contexts. *Right:* Run time increases with both K and N .

6.2 Comparing VI Parameterizations and Initialization Strategies

We compare Algorithm 1 (*base*) with updates assuming a correlated prior (*correlated*) and the full parameterization (Algorithm 2, *full*) across the experimental settings described in the previous section. We also explore the effect of initialization with factor analysis and varimax rotation, and the dropout gradient estimator. First we note that the factor analysis and varimax initialization improves algorithm performance in terms of log likelihood (Fig. 2, Supp. Fig. 3) and elementwise accuracy rate (Fig. 2, Supp. Fig. 4) across all experimental settings tested. Dropout gradient estimation tends to improve performance (under uniform random initialization) in terms of both log likelihood (Supp. Fig. 1) and EAR (Fig. 3). However, varimax initialization is shown to lead to superior performance to random initialization with dropout gradient estimation in terms of log likelihood (Fig. 4) and EAR (Supp Fig. 2).

Across the experimental settings tested, the *full* variational parameterization achieves higher likelihood values than the *base* approach, while the *correlated* prior achieves a marginal improvement over *base* (Fig. 7, Supp Fig. 5), especially in the $\rho > 0$ settings. However this improvement in log likelihood of the posterior mode does not translate into improved performance on EAR (Fig. 8, Supp Fig. 6). We speculate that the improved performance on

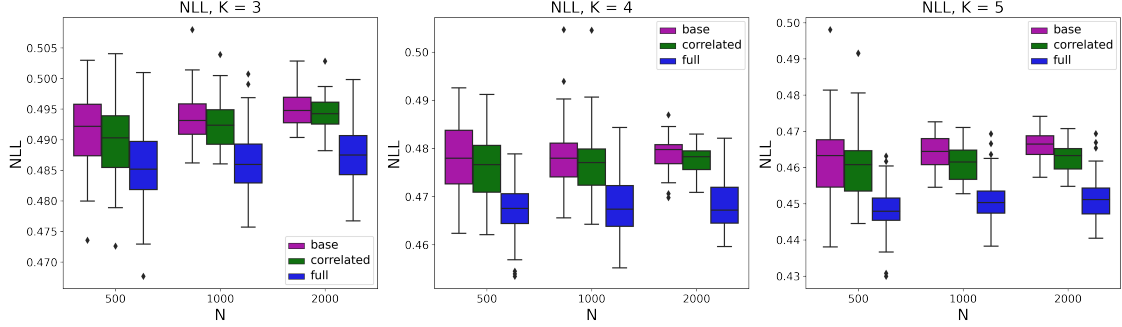


Figure 7: Negative log likelihood (NLL) comparison of variational inference parameterizations. NLL across a number of experimental settings evaluated at the VI posterior mode for the base DINA-CSP model, the DINA-CSP model with correlated prior and the DINA-CSP model with full parameterization of the latent skills A_i . Here, initialization is via varimax rotation, with $\rho = 0.25, s = 0.2, g = 0.2$.

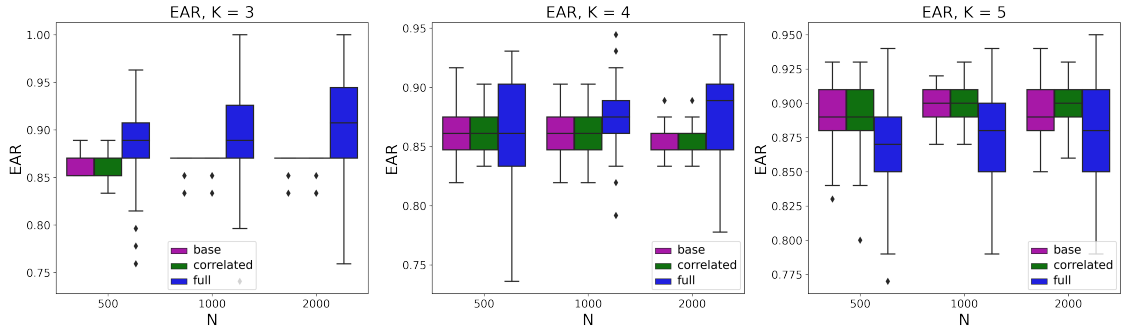


Figure 8: Elementwise accuracy rate (EAR) comparison of variational inference parameterizations. EAR across a number of experimental settings evaluated at the VI posterior mode for the base DINA-CSP model, the DINA-CSP model with correlated prior and the DINA-CSP model with full parameterization of the latent skills A_i . Here, initialization is via varimax rotation, with $\rho = 0.25, s = 0.2, g = 0.2$

NLL is due to the increased number of parameters in the alternative parameterizations, which mitigates the impact of pathological coordinate ascent fixed points in the ELBO objective. For high noise settings (i.e. $\rho > 0$) such in-sample likelihood improvements do not necessarily translate to accurate recovery of the ground truth parameters.

7 Real Data Applications

We fit DINA models on two large scale datasets in which exploratory item-attribute matrix estimation would be difficult with existing methods. Each of these datasets has a large

number of respondents as well as items and the number of latent components of the item-attribute matrix cannot be a priori determined.

7.1 TIMSS 2011 Student Response Data

We apply our algorithm to the TIMSS 2011 dataset containing Austrian student responses to a set of math questions. Previous works used subsets of the data for exploratory item-attribute matrix estimation with $N = 1010$ and $P = 47$; however, here we fit an exploratory DINA item-attribute matrix using the full dataset of 174 questions across 4668 students. Each student answered an average of 25 questions, leading to 115983 binary observations Y_{ij} for student i and question j . We initialized the item-attribute matrix using an expert-derived item-attribute matrix (Fig. 9, left).

The expert item-attribute matrix consists of 9 latent skills corresponding to the pairwise interactions between data, geometry, numbers and knowing, applying and reasoning. We fit a new DINA item-attribute matrix, setting $K_{\max} = 35$, and other parameters fixed to the values of the simulations. Our estimated $K = 9$. In total 5.5% of the original item-attribute matrix elements are modified. 3.3% involve an element of the item-attribute matrix moving from 1 to 0, while 2.2% involve an element of the item-attribute matrix moving from 0 to 1 (Fig. 9, middle right). *data reasoning* was the least modified latent skill (1.1% of item-attribute matrix entries) while *numerical knowing* was the most modified skill (12.6% of item-attribute matrix entries) (Fig. 9, right). A single question, M031185, is fully reclassified to a different latent skill - it is moved from NR to NA. This question involves converting distances on a map to their respective true distances by multiplying by a scale factor. It is reasonable that this question requires the *applying* cognitive domain.

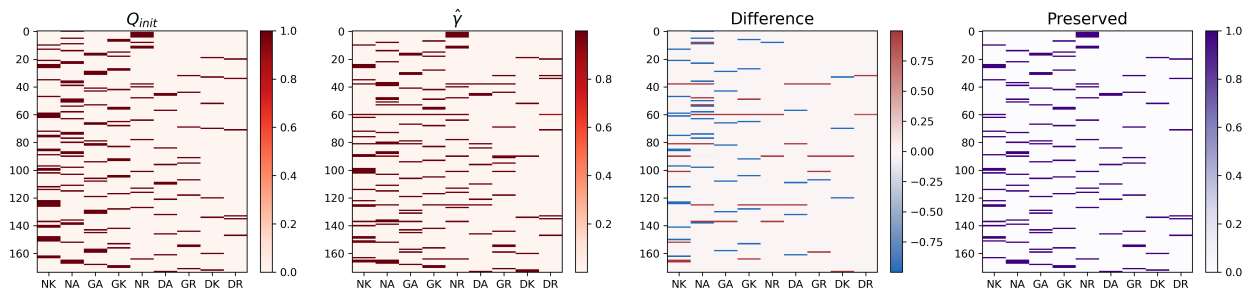


Figure 9: TIMSS 2011 item-attribute matrix after refinement by variational inference estimation algorithm. *left*: Original expert item-attribute matrix. *middle left*: Refined item-attribute matrix. *middle right*: Difference.

7.1.1 Items requiring multiple latent skills

The original item-attribute matrix has mutually exclusive components, meaning that each question requires only a single skill to answer. Hence there is a total of 174 positive entries for the 174 questions. After modification by the variational inference algorithm, 16 questions are found to require multiple skills (Table 4). These questions are generally the highest difficulty and have the fewest number of students answering them correctly (Fig. 10, left). These items are sensibly found to require additional skills. For example, item M041284 involves classifying shapes into a table based on the number of sides and whether the sides have the same length and is found to require GR and DR - the organization of data into a table makes this question mainly accessible to students that are proficient in the *data reasoning* skill. A geometry question, M031297, which involves computing the area of a shaded region is found to also require NA in addition to GA - possibly due to the computation with fractions required to arrive at the correct answer. Finally, M031016 involves writing all integers between 1 and 3000 that end in 112 and is found to require NR and NA skills.

Students with the highest number of latent skills, as measured by the estimate $\sum_{k=1}^K \alpha_{ik}$, score highest on the exam (Fig. 10, right).

	Co_NK	Co_NA	Co_GA	Co_GK	Co_NR	Co_DA	Co_GR	Co_DK	Co_DR
Item ID									
M041284	0	0	0	0	0	0	1	0	1
M051006	1	1	0	0	1	1	1	0	0
M031016	0	1	0	0	1	0	0	0	0
M031183	0	1	0	1	0	0	0	0	0
M031297	0	1	1	0	0	0	0	0	0
M041122	1	1	1	1	1	0	0	0	1
M031325	1	1	1	0	0	1	0	0	0
M051030	1	1	0	0	1	0	1	1	0
M041065B	1	0	0	0	1	0	1	0	0
M041059	1	1	0	1	1	1	0	0	0
M051075	1	1	1	0	1	0	0	0	0
M041165	0	0	1	0	0	0	1	0	0
M051008	0	0	0	1	1	0	0	0	0
M051031A	1	1	0	0	0	0	0	0	0
M051031B	1	1	0	0	0	0	0	0	0
M051507B	0	0	0	0	0	1	0	1	0

Table 4: 16 questions that were found to require multiple skills. **1** denotes a new required skill not found in the original item-attribute matrix. **1** denote required skills found in the original item-attribute matrix.

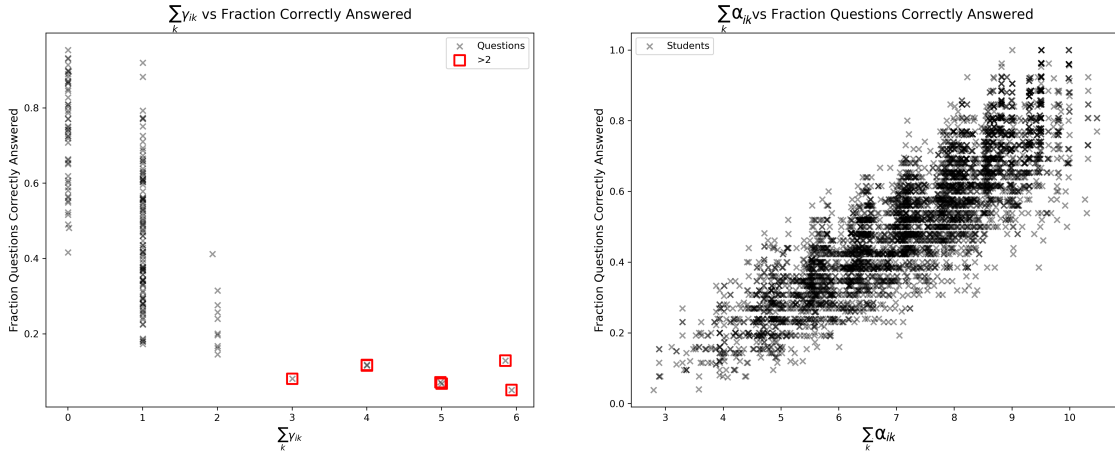


Figure 10: *left*: Fraction of students correctly answering question vs. the number of non-zero elements of the item-attribute matrix. Questions requiring multiple skills are generally harder. *right*: Fraction of questions correctly answered per student. The total number of skills a student has generally predicts their overall test score.

7.2 Single-cell ATACseq Data

To demonstrate the scalability of our approach, we applied our method to single cell assay for transposase accessible chromatin sequencing (ATACseq) data (Buenrostro et al. (2015),

Grandi et al. (2022)). Single cell ATACseq measures how "open" a region of DNA is via the ability of a hyperactive Tn5 transposase to insert sequencing adapters in the DNA. Functional regions of DNA (e.g. sequences acting as promoters or enhancers) can be identified via the enrichment of reads mapping to a particular stretch of the genome. Cell specific barcoding enables determining the context specificity of these functional regions. After processing the raw sequencing reads, common practice is to produce a cells by peak count matrix, which counts the number of reads per cell that coincide with each peak. Peaks are typically defined as 500 base pair segments of DNA with a high density of sequencing reads, as determined by a peak calling algorithm such as MACS2 (Gaspar (2018), Granja et al. (2021)).

To demonstrate the utility of our approach, we fit a DINO model to a single cell ATACseq dataset (Persad et al. (2023)) of CD34+ bone marrow cells collected from healthy human donors. These cells are the stem and progenitor cells that differentiate into various blood cell types, including erythrocytes, B cells, T cells, monocytes and dendritic cells (Fig. 11, b). Under the DINO model, the ideal response matrix entries become: $J_{ij} := 1 - \prod_{k=1}^K (1 - A_{ik} G_{jk})$. This model is mathematically equivalent to the DINA model after a certain reparameterization [YG: describe it here or in the Supplement] and can be fit by our VI algorithm. Using the preprocessed data of the original paper, we have a binarized count matrix of 6881 cells and 246113 peaks. Most peaks are non-informative, and so for each cell type we took the 16 most informative features, leading to a dataset with $N = 6881$ and $P = 122$. We fit the DINO model using CAVI-CSP with $K_{\max} = P$. The estimated latent dimension $\hat{K}$ is 5 (Fig. 11, c). The unique binary patterns have high agreement with the cell type annotations (Fig. 11, a-d). However, the latent binary patterns are more fine grained. For example, monocytes are represented by two binary patterns, 10100 and 00100, and HSCs are represented by two patterns 00001 and 00011. An interpretable feature of the result is that we can track how related cell types are based on their binary patterns. HMPs and HSCs are the most stem-like cell populations, and are both found to express the 5th binary latent

(Fig. 11, d). Additionally, the 3rd binary latent is specifically expressed by dendritic cells and monocytes, and hence is specific to myeloid identity cells.

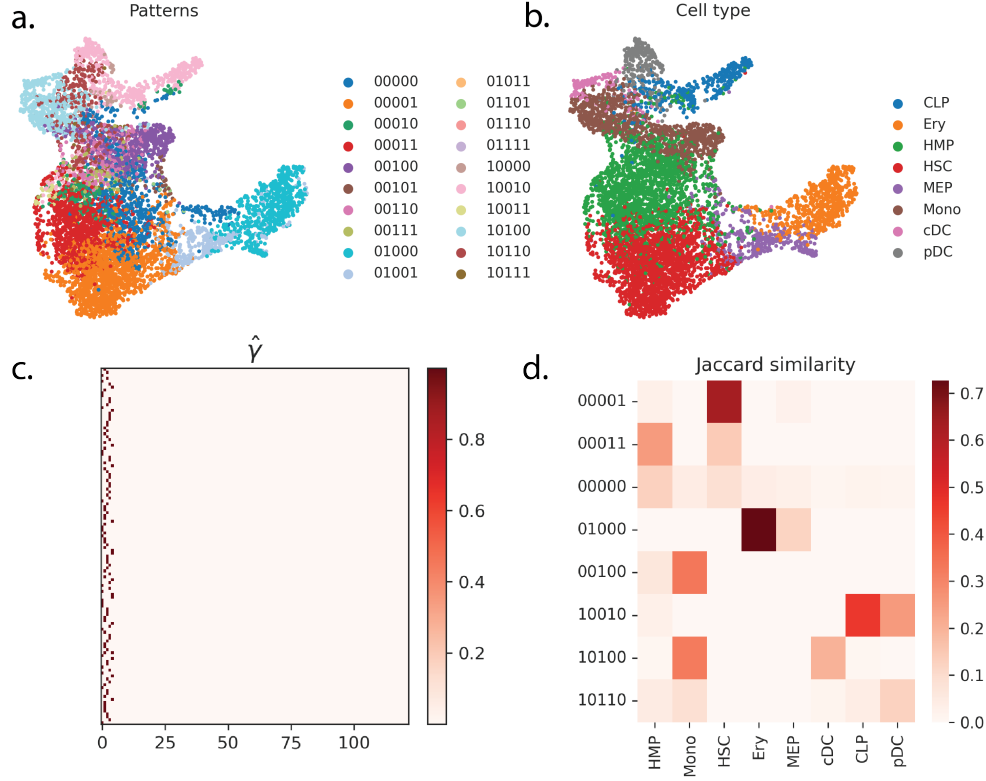


Figure 11: (a.) 2D projection (UMAP) of cells colored by latent binary pattern α_i . (b.) 2D projection (UMAP) of cells colored by cell type annotation (c.) Estimated item-attribute matrix, γ , showing the inferred dimension K is equal to 5. (d.) Pairwise Jaccard similarity between most common latent binary patterns and cell type labels.

8 Discussion

We have developed a scalable variational inference algorithm that estimates the parameters of the DINA model in the challenging scenario where the number of latent variables is unknown. Existing approaches require an exponential number of parameters in the number of latent variables K . We demonstrate empirically that a mean-field parameterization requiring $\mathcal{O}(K)$ parameters achieves competitive performance on this task, if properly initialized. Further, we establish the sample complexity of true parameters emerging as fixed points of

our algorithm, which occurs at a logarithmic rate. Empirically, we show that estimation of the number of latent components via the cumulative shrinkage prior achieves competitive performance with an existing MCMC-based method. Furthermore, proper initialization and stochastic updates improve algorithm performance. Two major underexplored directions are the optimization landscape of the evidence lower bound objective function and the behavior of spectral methods on data generated under the DINA model. Our work demonstrates that pathological fixed points of CAVI are prevalent, but we do not have a full understanding of their properties. Additionally, our work suggests that spectral methods can perform reasonably well at estimating the parameters of the DINA model, e.g. recover the sparsity pattern of the item-attribute matrix, but we do not know the exact conditions under which this is possible.

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Supplement to “Scalable Variational Inference for Boolean Matrix Factorization with an Unknown Latent Dimension”

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1 Technical Proofs

This section contains the technical proofs for the theoretical results in the main paper.

1.1 Proof of Proposition 1

Proposition 1 (Linear approximation to ideal response). *Let $A_{ik} \sim_{iid} \text{Bernoulli}(1/2)$, and let γ_N be defined as:*

$$\gamma_N = \underset{\gamma}{\operatorname{argmin}} \frac{1}{NP} \sum_{i=1}^N \sum_{j=1}^P \left(\prod_{k=1}^K (1 - (1 - A_{ik})G_{jk}) - \sum_{k=1}^K A_{ik}\gamma_{jk} \right)^2$$

Then $\gamma_N \rightarrow_{N \rightarrow \infty} \gamma^$ almost surely, with $\mathbf{1}\{\gamma_{jk}^* > 0\} = G_{jk}$ for all $j = 1, \dots, P$ and $k = 1, \dots, K$. For j such that $\sum_{k=1}^K G_{jk} = 1$, we have $\gamma_{jk}^* = G_{jk}$ for all $k \in [K]$.*

Proof. Define $\mathbf{J}$ as the matrix with $J_{ij} := \prod_{k=1}^K (1 - (1 - A_{ik})G_{jk})$. This follows from the regression formula:

$$\gamma_N^t = \left(\frac{1}{N} \mathbf{A}^\top \mathbf{A} \right)^{-1} \frac{1}{N} \mathbf{A}^\top \mathbf{J}$$

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Applying the law of large numbers and then the Woodbury matrix identity, we have:

$$\gamma_{jk}^* = 2 \left(\left(\frac{1}{2} \right)^{\sum_{l \neq k} G_{jl}} - \frac{1}{K+1} \sum_{l=1}^K \left(\frac{1}{2} \right)^{\sum_{m \neq l} G_{jm}} \right)$$

The result can be deduced from this expression. For a particular element, letting $L_j := \sum_{k=1}^K G_{jk}$, we have:

$$\gamma_{jk}^* = \left(\frac{1}{2} \right)^{L_j-1} \left(G_{jk} + 1 - \frac{K + L_j}{K + 1} \right)$$

we note that $1 \leq \frac{K+L_j}{K+1} < 2$, and so $\mathbf{1}\{\gamma_{jk}^* > 0\} = G_{jk}$.

□

1.2 Proof of Theorem 1

In Theorem 1 and Theorem 2, we assume $s_j = g_j =: s$ and $K = K_{\max}$. We hence use these notations interchangeably. We denote $\alpha_i \in \{0, 1\}^K$ as an estimate for the binary pattern of individual i , and $A_i \in \{0, 1\}^K$ as the true binary pattern. The following lemmas show that for sufficiently close initialization of the item-attribute matrix, the error of one step of coordinate ascent can be bounded in terms of P and N . These lemmas are necessary to prove Theorem 1.

Assumption 1. For all $(i, k) \in [N] \times [K]$

$$\sum_{j=1}^P G_{jk} \prod_{l \neq k} (1 - (1 - A_{il}) G_{jl}) = \Omega(P)$$

This states that for any individual and latent skill there are $\Omega(P)$ number of questions in the questionnaire that the individual can "almost answer" in the sense that they possess all latent skills besides the skill in question. Note that this is automatically satisfied when there are $\Omega(P)$ questions per skill that are unique to that skill. Further, for sufficiently large P let

:

$$\sum_{j=1}^P G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \geq \Xi P$$

for some $\Xi \in (0, 1)$.

Assumption 2. For all $(j, k) \in [P] \times [K]$

$$\sum_{i=1}^N (1 - A_{ik}) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = \Omega(N)$$

Further, for sufficiently large N let :

$$\sum_{i=1}^N (1 - A_{ik}) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \geq \Xi N$$

for some $\Xi \in (0, 1)$. This is the analogous assumption to Assumption 1, i.e. assuming adequate sampling for every item and latent across individuals.

The following lemma will be used in the proof of Lemmas 2 and 4.

Lemma 1 (Continuity of the ideal response differences). For $A \neq \alpha \in \{0, 1\}^K$, we first define the following ideal response differences:

$$D_{A,\alpha}(x) := \prod_{k=1}^K (1 - (1 - A_k)x_k) - \prod_{k=1}^K (1 - (1 - \alpha_k)x_k) : [0, 1]^K \rightarrow \mathbb{R}$$

for $x \in [0, 1]^K$. Then if $\|x - x^*\|_\infty \leq \rho \leq 1$ and $x^* \in \{0, 1\}^K$, we have $|D_{A,\alpha}(x) - D_{A,\alpha}(x^*)| \leq 2(1 - (1 - \rho)^K)$.

As a special case when A and α differ by a single element, i.e. $A_l = \alpha_l$ for all $l \neq k$, the above simplifies to:

$$D_{A,\alpha}(x) = (A_k - \alpha_k)x_k \prod_{l \neq k} (1 - (1 - A_l)x_l)$$

and

$$\left| x_k \prod_{l \neq k} (1 - (1 - A_l)x_l) - x_k^* \prod_{l \neq k} (1 - (1 - A_l)x_l^*) \right| \leq 2(1 - (1 - \rho)^K).$$

Proof. We start with triangle inequality:

$$\begin{aligned} & |D_{A,\alpha}(x) - D_{A,\alpha}(x^*)| \leq \\ & \left| \prod_{k=1}^K (1 - (1 - A_k)x_k) - \prod_{k=1}^K (1 - (1 - A_k)x_k^*) \right| + \left| \prod_{k=1}^K (1 - (1 - \alpha_k)x_k) - \prod_{k=1}^K (1 - (1 - \alpha_k)x_k^*) \right| \end{aligned}$$

So it suffices to find a bound for $|I_A(x^*) - I_A(x)|$ where $I_A(x)$ is the ideal response with A fixed: $I_A(x) := \prod_{k=1}^K (1 - (1 - A_k)x_k)$. We can apply the same bound for the first and second term.

First, we note that $I_A(x^*) \in \{0, 1\}$ under the assumptions, since each term of each product is either 0 or 1. So we can proceed by casework on $I_A(x^*)$.

Suppose $I_A(x^*) = 0$. Then there is at least one index k that has $A_k = 0$ and $x_k^* = 1$ (this is because we must have $(1 - A_k)x_k^* = 1$ for some k). By the assumption this means that $x_k \geq 1 - \rho$, and so $1 - (1 - A_k)x_k \leq \rho$. This implies $I_A(x) \leq \rho$, and so $|I_A(x) - I_A(x^*)| \leq \rho$.

Suppose $I_A(x^*) = 1$. This implies that $(1 - A_k)x_k^* = 0$ for all k . With additional casework, when $A_k = 1$, then $1 - (1 - A_k)x_k = 1 \geq 1 - \rho$. When $A_k = 0$, then we must have $x_k^* = 0$. So $x_k \leq \rho$. So:

$$(1 - (1 - A_k)x_k) \geq 1 - \rho$$

holds for all k , and we have $I_A(x) \geq (1 - \rho)^K$. $|I_A(x) - I_A(x^*)| \leq 1 - (1 - \rho)^K$.

The two cases establish $|I_A(x) - I_A(x^*)| \leq \max(1 - (1 - \rho)^K, \rho)$. If $\rho \in [0, 1]$, then we can simplify to $|I_A(x) - I_A(x^*)| \leq 1 - (1 - \rho)^K$ (since $(1 - \rho) \geq (1 - \rho)^K \implies \rho \leq 1 - (1 - \rho)^K$). This upper bound is free of A and so it also holds for α , which shows the conclusion:

$$|D_{A,\alpha}(x) - D_{A,\alpha}(x^*)| \leq 2(1 - (1 - \rho)^K)$$

□

The following lemma bounds the estimation error of students' skill pattern, measured by $\text{err}(\chi_i) := |1 - \chi_{i,A_i}|$, where A_i indexes into the true latent binary pattern for student i . For notation simplicity, we will use K in place of $K_{\max}$ in Algorithm 2.

Lemma 2 (Error bound for χ_i). *Assume that $\rho := \log(\frac{1-s_j}{g_j}) = \log(\frac{1-g_j}{s_j}) > 0$ with $\rho < \infty$ (This can be relaxed to allowing s_j and g_j to vary in a range). Consider the $2^K - 1$ events, one for each $\alpha_i \in \{0, 1\}^K$ and $\alpha_i \neq A_i$:*

$$\mathcal{E}_{\alpha_i} = \left\{ \left(\sum_{j=1}^P \rho(2Y_{ij} - 1) \left(\prod_{k=1}^K A_{ik}^{G_{jk}} - \prod_{k=1}^K \alpha_{ik}^{G_{jk}} \right) > Pc \right) \right\} \quad (1)$$

for a constant $c > 0$. For suitably chosen c , this event will occur with high probability. Then if $\mathcal{E}_{\alpha_i}$ holds for all α_i and we have an initialization such that $|\gamma_{jl} - G_{jl}| < \varrho$ for all (i, j, l) , after one CAVI update

$$\text{err}(\chi_i) \leq \exp \left(-P(c - 4\rho(1 - (1 - \varrho)^K)) \right) \quad (2)$$

Proof. With a slight overload of notation, we use χ_{i,A_i} to denote $\chi_{i,\sum_k A_{i,k} \cdot 2^k}$ by transforming the binary vector A_i from a base-2 number to its decimal representation. We note that:

$$\frac{\chi_{i,A_i}}{\chi_{i,\alpha_i}}$$

$$\begin{aligned}
&= \exp \left(\sum_{j=1}^P \left(\prod_{k=1}^K (1 - (1 - A_{ik})\gamma_{jk}) - \prod_{k=1}^K (1 - (1 - \alpha_{ik})\gamma_{jk}) \right) (Y_{ij} \log(\frac{1-s}{g}) - (1 - Y_{ij}) \log(\frac{1-g}{s})) \right) \\
&= \exp \left(\sum_{j=1}^P \rho(2Y_{ij} - 1) \left(\prod_{k=1}^K (1 - (1 - A_{ik})\gamma_{jk}) - \prod_{k=1}^K (1 - (1 - \alpha_{ik})\gamma_{jk}) \right) \right)
\end{aligned}$$

Denote $Q_\alpha := \prod_k A_{ik}^{G_{jk}} - \prod_k \alpha_{ik}^{G_{jk}}$, $\Delta_\alpha := \prod_k (1 - (1 - A_{ik})\gamma_{jk}) - \prod_k (1 - (1 - \alpha_{ik})\gamma_{jk}) - Q_\alpha$, we have

$$\frac{\chi_{i,A_i}}{\chi_{i,\alpha_i}} = \exp \left(\sum_{j=1}^P 2\rho(2Y_{ij} - 1)(Q_\alpha + \Delta_\alpha) \right)$$

We get the conclusion by applying Lemma 1 and under the events $\mathcal{E}_{\alpha_i}$:

$$\begin{aligned}
\frac{\chi_{i,A_i}}{\chi_{i,\alpha_i}} &\geq \exp \left(Pc + \sum_{j=1}^P 2\rho(2Y_{ij} - 1)\Delta_\alpha \right) \\
&\geq \exp \left(Pc - 4P\rho(1 - (1 - \rho)^K) \right) = \exp \left(P(c - 4\rho(1 - (1 - \rho)^K)) \right)
\end{aligned}$$

Then we have:

$$\begin{aligned}
\text{err}(\chi_i) &= 1 - \chi_{i,A_i} \\
&= \sum_{\alpha_i \neq A_i} \chi_{i,\alpha_i} \\
&= \chi_{i,A_i} \sum_{\alpha_i \neq A_i} \frac{\chi_{i,\alpha_i}}{\chi_{i,A_i}} \\
&\leq (2^K - 1) \exp \left(-P(c - 4\rho(1 - (1 - \rho)^K)) \right)
\end{aligned}$$

□

Lemma 3 (Concentration of event $\mathcal{E}_{\alpha_i}$). *Suppose $s < 1/2 - \epsilon$, $g < 1/2 - \epsilon$ and Assumption 1.*

Let $c := \rho \times \epsilon \times \Xi$ (Ξ as defined in Assumption 1, and ρ defined in Lemma 2). Then

$$\mathbb{P}\left[\bigcap_{\alpha_i} \mathcal{E}_{\alpha_i}\right] \geq 1 - N(2^K - 1) \exp\left(\frac{-P \times c}{\left(\frac{2}{3} + \frac{2}{\epsilon}\right)\rho}\right)$$

Proof. Recall that the event $\mathcal{E}_{\alpha_i}$ is defined for each $\alpha_i \neq A_i$ as

$$\mathcal{E}_{\alpha_i} = \left\{ \left(\sum_{j=1}^P \rho(2Y_{ij} - 1) \left(\prod_{k=1}^K A_{ik}^{G_{jk}} - \prod_{k=1}^K \alpha_{ik}^{G_{jk}} \right) > Pc \right\}.$$

Dropping the i subscript, define:

$$u_j := \rho(2Y_{ij} - 1) \left(\prod_{k=1}^K A_{ik}^{G_{jk}} - \prod_{k=1}^K \alpha_{ik}^{G_{jk}} \right)$$

Note that these variables are bounded so we aim to apply Bernstein's inequality. To show this inequality, some additional care is needed. We divide up the sum:

$$\begin{aligned} \sum_{j=1}^P u_j &= \sum_{j=1}^P \rho(2Y_{ij} - 1) \left(\prod_{k=1}^K A_{ik}^{G_{jk}} - \prod_{k=1}^K \alpha_{ik}^{G_{jk}} \right) \\ &= \sum_A \rho(2Y_{ij} - 1) - \sum_B \rho(2Y_{ij} - 1) \end{aligned}$$

where $A = \{j : \prod_k A_{ik}^{G_{jk}} = 1, \prod_k \alpha_{ik}^{G_{jk}} = 0\}$ and $B = \{j : \prod_k A_{ik}^{G_{jk}} = 0, \prod_k \alpha_{ik}^{G_{jk}} = 1\}$.

Note that on A , $Y_{ij} \sim \text{Bernoulli}(1 - s)$ and on B , $Y_{ij} \sim \text{Bernoulli}(g)$. Redefine $\tilde{Y}_{ij} := Y_{ij}$ for $j \in A$ and $\tilde{Y}_{ij} := 1 - Y_{ij}$ for $j \in B$. By assumption $s = g$, so for $\tilde{Y}_{ij} \sim_{\text{i.i.d}} \text{Bernoulli}(1 - s)$:

$$\sum_j u_j = \sum_{A \cup B} \rho(2\tilde{Y}_{ij} - 1)$$

We have shown:

$$D := \mathbb{E}[\sum_j u_j] = 2\rho(\frac{1}{2} - s)(|A| + |B|)$$

We now revisit the assumption:

$$\sum_{j=1}^P G_{jk} \prod_{l \neq k} A_{il}^{G_{jl}} \geq \Xi P$$

We deal with two cases separately: (1) there exists some k such that $A_{ik} > \alpha_{ik}$, (2) $\alpha_{ik} \geq A_{ik}$ for all $k \in [K]$. In the first case:

$$G_{jk} \prod_{l \neq k} A_{il}^{G_{jl}} = 1 \implies \prod_{k=1}^K A_{ik}^{G_{jk}} - \prod_{k=1}^K \alpha_{ik}^{G_{jk}} = 1$$

so that $|A| \geq \Xi P$, and $|A| + |B| \geq \Xi P$. In the other case, since $\alpha_i \neq A_i$, there is at least one k such that $\alpha_{ik} > A_{ik}$. We also note that $\prod_{l \neq k} A_{il}^{G_{jl}} = 1 \implies \prod_{l \neq k} \alpha_{il}^{G_{jl}} = 1$, and so:

$$G_{jk} \prod_{l \neq k} A_{il}^{G_{jl}} = 1 \implies G_{jk} \prod_{l \neq k} \alpha_{il}^{G_{jl}} = 1 \implies \prod_{k=1}^K A_{ik}^{G_{jk}} - \prod_{k=1}^K \alpha_{ik}^{G_{jk}} = -1$$

so that $|B| \geq \Xi P$ and $|A| + |B| \geq \Xi P$. Thus we have $D \geq 2\rho\epsilon\Xi P = 2Pc$.

Note that $u_j \in \{-\rho, \rho, 0\}$ almost surely. The second moment can thus be computed directly as:

$$\begin{aligned} \sum_j \mathbb{E}[u_j^2] &= \rho^2 \times (|A| + |B|) \\ \sum_{j=1}^P \mathbb{E}[u_j^2] &= \frac{\rho D}{2(\frac{1}{2} - s)} \end{aligned}$$

Applying Bernstein's inequality (Vershynin, 2018) then gives:

$$\begin{aligned}
\mathbb{P}\left(\sum_{j=1}^P u_j \geq \frac{D}{2}\right) &= \mathbb{P}\left(\sum_{j=1}^P u_j - D \geq -\frac{D}{2}\right) \\
&= 1 - \mathbb{P}\left(\sum_{j=1}^P u_j - D < -\frac{D}{2}\right) \\
&\geq 1 - \mathbb{P}\left(\left|\sum_{j=1}^P u_j - D\right| \geq \frac{D}{2}\right) \\
&\geq 1 - \exp\left(-\frac{\frac{1}{2}\left(\frac{D}{2}\right)^2}{\sum_{j=1}^P \text{Var}(u_j) + \frac{1}{6}\rho D}\right) \\
&\geq 1 - \exp\left(-\frac{\frac{1}{2}\left(\frac{D}{2}\right)^2}{\sum_{j=1}^P \mathbb{E}(u_j^2) + \frac{1}{6}\rho D}\right) \\
&= 1 - \exp\left(-\frac{\frac{1}{2}\left(\frac{D}{2}\right)^2}{\frac{D\rho}{2(\frac{1}{2}-s)} + \frac{1}{6}\rho D}\right) \\
&= 1 - \exp\left(-\frac{D}{\left(\frac{4}{3} + \frac{4}{(\frac{1}{2}-s)}\right)\rho}\right) \\
&\geq 1 - \exp\left(-\frac{D}{\left(\frac{4}{3} + \frac{4}{\epsilon}\right)\rho}\right) \\
&\geq 1 - \exp\left(-\frac{2Pc}{\left(\frac{4}{3} + \frac{4}{\epsilon}\right)\rho}\right)
\end{aligned}$$

Finally we note $D/2 \geq Pc$, so

$$\mathbb{P}\left(\sum_{j=1}^P u_j \geq Pc\right) \geq 1 - \exp\left(-\frac{Pc}{\left(\frac{2}{3} + \frac{2}{\epsilon}\right)\rho}\right)$$

Applying the union bound shows the desired result. $\square$

Next, we provide a lemma for γ_{jk} , for which the analysis is similar, but we include it for completeness.

Lemma 4 (Error bound for γ_{jk}). *Let $\text{err}(\gamma_{jk}) = |\gamma_{jk} - G_{jk}|$. Define ρ as in Lemma 2.*

Consider the event

$$\mathcal{E}' = \left\{ (2G_{jk} - 1) \left(\sum_{i=1}^N (A_{ik} - 1) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \times \rho(2Y_{ij} - 1) \right) > Pc \right\} \quad (3)$$

Then if $\mathcal{E}'$ holds and we have an initialization such that $|\alpha_{il} - A_{il}| < \varrho$, $|\gamma_{jl} - G_{jl}| < \varrho$ for all (i, j, l) , after one CAVI update:

$$\text{err}(\gamma_{jk}) \leq \mathcal{C}(\delta) \exp \left(-N(c - \rho(1 - (1 - \varrho)^K)) \right)$$

where $\mathcal{C}(\delta) = \frac{1-\delta}{\delta}$, $\alpha_i = \text{bin}_K(\arg \max(\chi_i))$.

Proof. Under the assumptions, the update for γ_{jk} is given by:

$$\gamma_{jk} = \sigma \left(\sum_{i=1}^N \rho(2Y_{ij} - 1) \times (\alpha_{ik} - 1) \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) + \left(\sum_{m=1}^k \phi_{km} \log \left(\frac{\delta}{1 - \delta} \right) \right) \right)$$

Applying the same strategy as in Lemma 1, we define:

$$\Delta_i := (\alpha_{ik} - 1) \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) - (A_{ik} - 1) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl})$$

and write the update as

$$\gamma_{jk} = \sigma \left(\sum_{i=1}^N \rho(2Y_{ij} - 1) \times \left((A_{ik} - 1) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) + \Delta_i \right) + \left(\sum_{m=1}^k \phi_{km} \log \left(\frac{\delta}{1 - \delta} \right) \right) \right)$$

Similar casework argument as in Lemma 2 can be applied for achieving $|\Delta_i| \leq 1 - (1 - \varrho)^K$.

First we consider the case $(A_{ik} - 1) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = 0$. If $A_{ik} = 1$, then we must have $\alpha_{ik} \leq \varrho$, and it follows $|(\alpha_{ik} - 1) \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl})| \leq \varrho$. Otherwise we have $\prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = 0$, which implies that $A_{im} = 0$ and $G_{jm} = 1$ for some $m \neq k$. This implies $\alpha_{im} \leq \varrho$, $\gamma_{jk} \geq 1 - \varrho$, from which we can establish $1 - (1 - \alpha_{im})\gamma_{jm} \leq 1 - (1 - \varrho)^2$.

It follows that $|(\alpha_{ik} - 1) \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl})| \leq 1 - (1 - \varrho)^2$. Finally we consider the case

with $(A_{ik} - 1) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = -1$. We have $A_{ik} = 0$, and so $\alpha_{ik} \leq \varrho$. If $G_{jl} = 0$, then $\gamma_{jl} \leq \varrho$ which implies $(1 - \alpha_{jl})\gamma_{jl} \leq \gamma_{jl} \leq \varrho$. This establishes $1 - (1 - \alpha_{jl})\gamma_{jl} \geq 1 - \varrho$. If $G_{jl} = 1$ then $A_{jl} = 1$ and

$$\alpha_{jl} \geq 1 - \varrho$$

$$1 - \alpha_{jl} \leq \varrho$$

$$(1 - \alpha)\gamma_{jl} \leq \varrho$$

Each term in $(1 - \alpha_{ik}) \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl})$ is greater than or equal to $1 - \varrho$ and so $(1 - \alpha_{ik}) \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \geq (1 - \varrho)^K$. Combining results we have :

$$|\Delta_i| \leq \max(1 - (1 - \varrho)^K, 1 - (1 - \varrho)^2, \varrho)$$

We proceed by casework on G_{jk} . For $G_{jk} = 1$, we have under event $\mathcal{E}'$:

$$\sum_{i=1}^N (A_{ik} - 1) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \times \rho(2Y_{ij} - 1) > Nc$$

And so we have the error bound:

$$\begin{aligned} \text{err}(\gamma_{jk}) &\leq \exp \left(- \sum_{i=1}^N \rho(2Y_{ij} - 1) \times \left((A_{ik} - 1) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) + \Delta_i \right) - \left(\sum_{m=1}^k \phi_{km} \right) \log \left(\frac{\delta}{1 - \delta} \right) \right) \\ &\leq \exp \left(- Nc - \sum_{i=1}^N \rho(2Y_{ij} - 1) \Delta_i + \log \left(\frac{1 - \delta}{\delta} \right) \right) \\ &\leq \mathcal{C}(\delta) \exp \left(- Nc + N\rho(1 - (1 - \varrho)^K) \right) \end{aligned}$$

We now consider the case $G_{jk} = 0$, where we have under event $\mathcal{E}'$:

$$- \sum_{i=1}^N (A_{ik} - 1) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \times \rho(2Y_{ij} - 1) > Nc$$

The error can be bounded as:

$$\begin{aligned}
err(\gamma_{jk}) &\leq \exp \left(\sum_{i=1}^N \rho(2Y_{ij} - 1) \times \left((A_{ik} - 1) \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) + \Delta_i \right) + \left(\sum_{m=1}^k \phi_{km} \right) \log \left(\frac{\delta}{1 - \delta} \right) \right) \\
&\leq \exp \left(-Nc + \sum_{i=1}^N \rho(2Y_{ij} - 1)\Delta_i \right) \\
&\leq \exp \left(-Nc + N\rho(1 - (1 - \varrho)^K) \right)
\end{aligned}$$

This completes the proof, as $1 \leq \frac{1-\delta}{\delta}$. □

Lemma 5 (Concentration of event $\mathcal{E}'$). *We require the identifying assumptions $s < 1/2 - \epsilon$, $g < 1/2 - \epsilon$ and Assumption 2. Namely, let $c := \rho \times \epsilon \times \Xi$ (Ξ as defined in Assumption 2, and ρ defined in Lemma 1). Then*

$$\mathbb{P}[\mathcal{E}'] \geq 1 - PK \exp \left(\frac{-N \times c}{(\frac{2}{3} + \frac{2}{\epsilon})\rho} \right)$$

Proof. The same steps as in Lemma 3 can be applied. □

These results can be combined for a statement about the performance of CAVI-CSP with sufficiently close initialization.

Theorem 1 (Sample complexity of true parameters being fixed points). *We first define some fundamental constants that determine the difficulty of the problem setting. For simplicity of notation, we let $s_j = g_j = s$ for all $j = 1, \dots, P$ and introduce the assumption $\epsilon := \frac{1}{2} - s > 0$. ϵ quantifies one aspect of the problem difficulty, namely the level of noise in the problem. Additionally, let $\rho := \log(\frac{1-s}{g}) > 0$, which is a function of ϵ under the assumptions. Further, let Ξ be the constant defined in Assumption 1 and Assumption 2.*

Suppose for all $(i, j, k) \in [N] \times [P] \times [K]$:

$$|\gamma_{ik} - G_{ik}| \leq 1 - (1 - \frac{\epsilon\Xi}{2})^{1/K}.$$

Then for estimates obtained via a finite number of iterations of CAVI-CSP, with probability at least $1 - \xi$, for all $(i, j, k) \in [N] \times [P] \times [K]$:

$$|\gamma_{jk} - G_{jk}| < \mathcal{M}$$

$$m_i := \arg \max(\chi_i) = A_i$$

when the number of items and respondents satisfy:

$$N = \tilde{\Omega}(\frac{1}{\epsilon\Xi \min(\epsilon, \rho)}), \quad P = \tilde{\Omega}(\frac{1}{\epsilon\Xi \min(\epsilon, \rho)})$$

Proof. We will apply Lemmas 2 to 5 to establish the sample complexity above. There are three conditions necessary to ensure that α and γ are fixed points of the coordinate ascent algorithm, which all depend on sample size P and N . Intuitively these are (1) the iterates form a contraction so that the assumptions of Lemmas 2 and 4 hold inductively, (2) the error bounds in Lemmas 2 and 4 are each less than $1/2$ and (3) the events $\mathcal{E}$ and $\mathcal{E}'$ hold with sufficiently high probability. We divide the proof into these three sections:

Part 1: Suppose the event $\mathcal{E} \cap \mathcal{E}'$ hold. And suppose that P is sufficiently large so that:

$$P\rho \geq \frac{\log\left(\frac{1}{1-(1-\epsilon\Xi/2)^{1/K}}\right)}{\epsilon\Xi/2}. \quad (4)$$

Multiplying by -1 on both sides of the inequality and exponentiating gives

$$\exp\left(-P\rho\frac{\epsilon\Xi}{2}\right) \leq 1 - (1 - \frac{\epsilon\Xi}{2})^{1/K}. \quad (5)$$

Applying Lemma 2 with $\varrho = 1 - (1 - \frac{\epsilon\Xi}{2})^{1/K}$ shows that

$$\text{err}(\chi_i^{(1)}) \leq \exp\left(-P\rho\frac{\epsilon\Xi}{2}\right) \quad (6)$$

for the first iteration. Since $1 - (1 - \frac{\epsilon\Xi}{2})^{1/K} \leq 1/2$ under the assumptions, this establishes for $m_i^{(t)} := \text{argmax}(\chi_i^{(t)})$, $\alpha_i^{(t)} := \text{bin}_K(m_i^{(t)}) = A_i$ for $t = 1$. In other words, we recover the correct pattern for A by taking the mode after the first update of CAVI.

Suppose N is large enough so that:

$$N\rho \geq \frac{\log\left(\frac{\mathcal{C}(\delta)}{1 - (1 - \frac{\epsilon\Xi}{2})^{1/K}}\right)}{\epsilon\Xi/2} \quad (7)$$

with $\mathcal{C}(\delta) = (1 - \delta)/\delta$. Then multiplying by -1 on both sides of the inequality and exponentiating gives:

$$\mathcal{C}(\delta) \exp\left(-N\rho\frac{\epsilon\Xi}{2}\right) \leq 1 - (1 - \frac{\epsilon\Xi}{2})^{1/K}. \quad (8)$$

Applying Lemma 4 by setting $\varrho = 1 - (1 - \frac{\epsilon\Xi}{2})^{1/K}$, $c = \rho\epsilon\Xi$, and with $|\alpha_i^{(1)} - A_i| = 0$, it gives the bound on the error:

$$\text{err}(\gamma_{jk}^{(1)}) \leq \mathcal{C}(\delta) \exp\left(-N\rho\frac{\epsilon\Xi}{2}\right) \leq \varrho. \quad (9)$$

Eq. (9) allows for the application of Lemma 2 to bound the error of the next step $\alpha_i^{(2)}$, which further allows for the application of Lemma 4 to bound the error of $\gamma_j^{(2)}$. Hence, recursively, for all iterations t ,

$$\text{err}(\gamma_{jk}^{(t)}) \leq \mathcal{C}(\delta) \exp\left(-N\rho\frac{\epsilon\Xi}{2}\right) \leq \varrho. \quad (10)$$

Part 2: To ensure $\text{err}(\gamma_{jk}^{(t)}) \leq \mathcal{M}$, it is sufficient to show

$$\mathcal{C}(\delta) \exp(-N\rho\epsilon\Xi/2) < \mathcal{M}, \quad (11)$$

for which it suffices to have:

$$N\rho > \frac{2 \log \frac{1}{\mathcal{M}}(\mathcal{C}(\delta))}{\epsilon\Xi}. \quad (12)$$

Part 3: We note that by Lemmas 3 and 4, the event $\mathcal{E} \cap \mathcal{E}'$ occurs with probability at least:

$$\mathbb{P}[\mathcal{E} \cap \mathcal{E}'] \geq 1 - N(2^K - 1) \exp\left(\frac{-P \times \epsilon\Xi}{\frac{2}{3} + \frac{2}{\epsilon}}\right) - PK \exp\left(\frac{-N \times \epsilon\Xi}{\frac{2}{3} + \frac{2}{\epsilon}}\right) \quad (13)$$

We need to choose N and P sufficiently large such that this is at least $1 - \xi$. One choice is choosing N and P to satisfy:

$$\begin{aligned} P &\geq \frac{\frac{2}{3} + \frac{2}{\epsilon}}{\epsilon\Xi} \log\left(\frac{2N(2^K - 1)}{\xi}\right) \\ N &\geq \frac{\frac{2}{3} + \frac{2}{\epsilon}}{\epsilon\Xi} \log\left(\frac{2PK}{\xi}\right) \end{aligned} \quad (14)$$

N and P satisfying each of these conditions is sufficient for Theorem 1, and hence we determine the form of F_1 and F_2 . That is, it suffices to take:

$$\begin{aligned} N &\geq \frac{1}{\epsilon\Xi} \left(\frac{2}{\rho} \log\left(\frac{\mathcal{C}(\delta)}{1 - (1 - \epsilon\Xi/2)^{1/K}}\right) + \frac{2}{\rho} \log\left(\frac{1}{\mathcal{M}}\mathcal{C}(\delta)\right) + \left(\frac{2}{3} + \frac{2}{\epsilon}\right) \log\left(\frac{2PK}{\xi}\right) \right) \\ P &\geq \frac{1}{\epsilon\Xi} \left(\frac{2}{\rho} \log\left(\frac{1}{1 - (1 - \epsilon\Xi/2)^{1/K}}\right) + \left(\frac{2}{3} + \frac{2}{\epsilon}\right) \log\left(\frac{2N(2^K - 1)}{\xi}\right) \right) \end{aligned} \quad (15)$$

Summarizing, we have:

$$\begin{aligned} N &= \tilde{\Omega}\left(\frac{1}{\epsilon\Xi \min(\epsilon, \rho)}\right) \\ P &= \tilde{\Omega}\left(\frac{1}{\epsilon\Xi \min(\epsilon, \rho)}\right) \end{aligned} \tag{16}$$

□

1.3 Proof of Theorem 2

Theorem 2 (Minimax lower bound). *We show that $\epsilon\rho\Xi$ is the fundamental quantity that determines the sample complexity. Let the parameter space Θ be the set of $\mathbf{A}$ and $\mathbf{G}$ such that Assumption 1 and Assumption 2 hold (with constant Ξ). Then as long as Θ is non-empty:*

- If $P \leq \kappa \frac{1}{\epsilon\rho\Xi}$ for $0 < \kappa < 1$, then:

$$\inf_{\hat{\mathbf{A}}} \sup_{\mathbf{A}, \mathbf{G} \in \Theta} \mathbb{P}(\hat{\mathbf{A}} \neq \mathbf{A}) \geq \frac{1}{4}(1 - \sqrt{\kappa})$$

- If $N \leq \kappa \frac{1}{\epsilon\rho\Xi}$ for $0 < \kappa < 1$, then:

$$\inf_{\hat{\mathbf{G}}} \sup_{\mathbf{A}, \mathbf{G} \in \Theta} \mathbb{P}(\hat{\mathbf{G}} \neq \mathbf{G}) \geq \frac{1}{4}(1 - \sqrt{\kappa})$$

Proof. We prove the first assertion (the second one is an equivalent argument). We lower bound the minimax risk via Le Cam's two point method. We have $\mathbb{P}(\hat{\mathbf{A}} \neq \mathbf{A}) \geq \mathbb{P}(\hat{A}_{ik} \neq A_{ik})$ for any choice of (i, k) so it suffices to find an (i, k) such that the bound holds. Intuitively, we would like to choose two points $(\mathbf{A}, \mathbf{G})$ and $(\mathbf{A}', \mathbf{G}')$ with $A_{ik} \neq A'_{ik}$ such that $D_{\text{KL}}(P(\mathbf{Y}|\mathbf{A}, \mathbf{G}), P(\mathbf{Y}|\mathbf{A}', \mathbf{G}'))$ is as small as possible. Under the constraint of assumption 1 we must have $\sum_{j=1}^P G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \geq \Xi P$. Without loss of generality we choose $\mathbf{A}, \mathbf{G}$ and $\mathbf{A}', \mathbf{G}'$ such that $A_{i'k'} = A'_{i'k'}$ for all (i', k') except (i, k) and $G = G'$, and further

such that $\sum_{j=1}^P G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = \Xi P^\dagger$. It follows that:

$$\begin{aligned}
D_{\text{KL}}(P(\mathbf{Y}|\mathbf{A}, \mathbf{G}), P(\mathbf{Y}|\mathbf{A}', \mathbf{G}')) &= \sum_{j=1}^P D_{\text{KL}}(P(Y_{ij}|\mathbf{A}, \mathbf{G}), P(Y_{ij}|\mathbf{A}', \mathbf{G}')) \\
&= \Xi P D_{\text{KL}}(\text{Bern}(1-s), \text{Bern}(g)) \\
&= \Xi P \left(\log \frac{s}{1-g} + (1-s) \left[\log \frac{1-s}{g} + \log \frac{1-g}{s} \right] \right) \\
&= 2\epsilon \rho \Xi P
\end{aligned}$$

Applying Le Cam's method with the 0 – 1 loss and Pinsker's inequality we get:

$$\begin{aligned}
\inf_{\hat{\mathbf{A}}} \sup_{\mathbf{A}, \mathbf{G} \in \Theta} \mathbb{P}(\hat{\mathbf{A}} \neq \mathbf{A}) &\geq \frac{1}{4} (1 - \text{TV}(P(\mathbf{Y}|\mathbf{A}, \mathbf{G}), P(\mathbf{Y}|\mathbf{A}', \mathbf{G}')) \\
&\geq \frac{1}{4} \left(1 - \sqrt{\frac{1}{2} D_{\text{KL}}(P(\mathbf{Y}|\mathbf{A}, \mathbf{G}), P(\mathbf{Y}|\mathbf{A}', \mathbf{G}'))} \right) \\
&\geq \frac{1}{4} \left(1 - \sqrt{\epsilon \rho \Xi P} \right) \\
&\geq \frac{1}{4} \left(1 - \sqrt{\kappa} \right)
\end{aligned}$$

Since $P \leq \kappa \frac{1}{\epsilon \rho \Xi}$, we have shown the assertion.

[†]It is clear that we can choose $\sum_{j=1}^P G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = \lceil \Xi P \rceil$. To see this, take any $\mathbf{A}$ and $\mathbf{G}$ that satisfy Assumption 1 and Assumption 2. Suppose $\min_{i,k} \sum_{j=1}^P G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) > \lceil \Xi P \rceil$. Then we also have $\sum_{j=1}^P G_{jk} > \lceil \Xi P \rceil$. For questions j such that $G_{jk} = 1$ and $\prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = 1$ we can construct a new $\tilde{\mathbf{G}}$ by setting G_{jk} to 0 successively until $\sum_{j=1}^P \tilde{G}_{jk} \prod_{l \neq k} (1 - (1 - A_{il})\tilde{G}_{jl}) = \lceil \Xi P \rceil$. It is easily verified that none of the other constraints are affected. $\square$

2 Derivations for variational inference

In this section, we derive the coordinate updates of Algorithm 1. These are found via first order conditions of the ELBO. To reduce notational burden, we understand each update to be in terms of the parameter values for iteration $t - 1$ without explicitly specifying this. We also understand K in this section to be $K_{\max}$ as the true value of K is unknown.

2.1 Update for γ_{jk}

Taking derivative of the ELBO with respect to γ_{jk} gives:

$$\begin{aligned} & \sum_{i=1}^N \left(- (1 - \alpha_{ik}) \prod_{l \neq k} (1 - (1 - \alpha_{il}) \gamma_{jl}) \right) \psi_{1,j}(Y_{ij}) + \left((1 - \alpha_{ik}) \prod_{l \neq k} (1 - (1 - \alpha_{il}) \gamma_{jl}) \right) \psi_{2,j}(Y_{ij}) \\ & + \left(\sum_{l=1}^k \phi_{kl} \right) \log\left(\frac{\delta}{1 - \delta}\right) + \log(1 - \gamma_{jk}) - \log(\gamma_{jk}) \end{aligned}$$

Note that the last two terms are the only ones that have a dependence on γ_{jk} . Setting to 0 and solving for γ_{jk} gives the update:

$$\begin{aligned} \gamma_{jk} & \leftarrow \sigma \left(\sum_{i=1}^N \left(- (1 - \alpha_{ik}) \prod_{l \neq k} (1 - (1 - \alpha_{il}) \gamma_{jl}) \right) \psi_{1,j}(Y_{ij}) + \left((1 - \alpha_{ik}) \prod_{l \neq k} (1 - (1 - \alpha_{il}) \gamma_{jl}) \right) \psi_{2,j}(Y_{ij}) \right. \\ & \quad \left. + \left(\sum_{l=1}^k \phi_{kl} \right) \log\left(\frac{\delta}{1 - \delta}\right) \right) \\ & = \sigma \left(\nabla_{\gamma_{jk}} \tilde{L}(\alpha, \gamma) \right) \end{aligned}$$

Note that $\log(\frac{\delta}{1 - \delta})$ is generally a negative number with large magnitude. So γ_{jk} tends to be small when $\sum_{l=1}^k \phi_{kl}$ is large. In addition, the update $\sigma(\nabla_{\gamma_{jk}} \tilde{L}(\alpha, \gamma))$ is equal to

$\sigma(\tilde{L}(\alpha, \gamma; \gamma_{jk} = 1) - \tilde{L}(\alpha, \gamma; \gamma_{jk} = 0))$, which indicates that

$$\mathbf{1}\{\sigma(\tilde{L}(\alpha, \gamma; \gamma_{jk} = 1) - \tilde{L}(\alpha, \gamma; \gamma_{jk} = 0)) > 0.5\} = \mathbf{1}\{\tilde{L}(\alpha, \gamma; \gamma_{jk} = 1) > \tilde{L}(\alpha, \gamma; \gamma_{jk} = 0)\}.$$

So, the rounded CAVI updates are equivalent to coordinate ascent updates directly on the log-likelihood.

2.2 Update for α_{ik}

Taking derivative of the ELBO with respect to α_{ik} gives:

$$\begin{aligned} & \sum_{j=1}^P \left(\gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \right) \psi_{1,j}(Y_{ij}) + \left(-\gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \right) \psi_{2,j}(Y_{ij}) \\ & + \log(1 - \alpha_{ik}) - \log(\alpha_{ik}) \end{aligned}$$

Setting to 0 and solving for α_{ik} gives the update:

$$\begin{aligned} \alpha_{ik} & \leftarrow \sigma \left(\sum_{j=1}^P \left(\gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \right) \psi_{1,j}(Y_{ij}) + \left(-\gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \right) \psi_{2,j}(Y_{ij}) \right) \\ & = \sigma \left(\nabla_{\alpha_{ik}} \tilde{L}(\alpha, \gamma) \right) \end{aligned}$$

2.3 Updates for s and g

Note that $\phi_{1,j}(Y_{ij})$ is the only term that depends on s_j . Taking the gradient we have:

$$\sum_{i=1}^N \left(\prod_{k=1}^K (1 - (1 - \alpha_{ik})\gamma_{jk}) \left(\frac{Y_{ij}}{1 - s_j} - \frac{1 - Y_{ij}}{s_j} \right) \right) = 0$$

Simplifying leads to the update

$$s_j = \frac{\sum_{i=1}^N (1 - Y_{ij}) \left(\prod_{k=1}^K (1 - (1 - \alpha_{ik}) \gamma_{jk}) \right)}{\sum_{i=1}^N \prod_{k=1}^K (1 - (1 - \alpha_{ik}) \gamma_{jk})}$$

Note that $\phi_{0,j}(Y_{ij})$ is the only term that depends on g_j . Taking the gradient we have:

$$\sum_{i=1}^N \left(\left(1 - \prod_{k=1}^K (1 - (1 - \alpha_{ik}) \gamma_{jk}) \right) \left(\frac{Y_{ij}}{g_j} - \frac{1 - Y_{ij}}{1 - g_j} \right) \right) = 0$$

Simplifying leads to the update

$$g_j = \frac{\sum_{i=1}^N Y_{ij} \left(1 - \prod_{k=1}^K (1 - (1 - \alpha_{ik}) \gamma_{jk}) \right)}{\sum_{i=1}^N \left(1 - \prod_{k=1}^K (1 - (1 - \alpha_{ik}) \gamma_{jk}) \right)}$$

2.4 Updates for ϕ

There are two cases to consider: $l \geq k + 1$ and $l \leq k$. For the first case, taking the derivative of the ELBO with respect to ϕ_{kl} gives :

$$\nabla_{\phi_{kl}} \text{ELBO} = \sum_{j=1}^P \log\left(\frac{1}{2}\right) + \mathbb{E}_Q \left[\nu_l \prod_{m=1}^{l-1} (1 - \nu_m) \right] - 1 - \log(\phi_{kl})$$

For $l \leq k$, the gradient is:

$$\nabla_{\phi_{kl}} \text{ELBO} = \sum_{j=1}^P (\gamma_{jk} \log \delta + (1 - \gamma_{jk}) \log(1 - \delta)) + \mathbb{E}_Q \left[\nu_l \prod_{m=1}^{l-1} (1 - \nu_m) \right] - 1 - \log(\phi_{kl})$$

Setting to 0 and rearranging these expressions, applying the constraint that $\sum_{l=1}^K \phi_{kl} = 1$, we arrive at the updates:

$$\phi_{kl} \propto \begin{cases} \exp \left(\mathbb{E}_\pi(\log \omega_l) + P \log(1/2) \right) & \text{if } l \geq k + 1 \\ \exp \left(\mathbb{E}_\pi(\log \omega_l) + \sum_{j=1}^P \gamma_{j,k} \log(\delta) + (P - \sum_{j=1}^P \gamma_{j,k}) \log(1 - \delta) \right) & \text{if } l < k + 1 \end{cases}$$

2.5 Updates for π

Recall that the variational posterior has $\nu_l \sim \text{Beta}(\pi_{0l}, \pi_{1l})$. The terms of the ELBO depending on ν_l are:

$$\sum_{k=1}^K \sum_{l=1}^K \phi_{kl} \left(\mathbb{E}_Q[\log \nu_l] + \sum_{m=1}^{l-1} \mathbb{E}_Q[\log(1 - \nu_m)] \right) + (\kappa - 1) \mathbb{E}_Q[\log(1 - \nu_l)] + H_Q$$

Using the result for Beta distributions $\mathbb{E}[\log \nu_l] = \Psi(\pi_{0l}) - \Psi(\pi_{0l} + \pi_{1l})$ and the entropy is:

$$H_Q = \log \Gamma(\pi_{0l}) + \log \Gamma(\pi_{1l}) - \log \Gamma(\pi_{0l} + \pi_{1l}) \quad (17)$$

$$- (\pi_{0l} - 1) \Psi(\pi_{0l}) - (\pi_{1l} - 1) \Psi(\pi_{1l}) + (\pi_{0l} + \pi_{1l} - 2) \Psi(\pi_{0l} + \pi_{1l}) \quad (18)$$

where Ψ is the digamma function and Γ is the gamma function. The full expression is then:

$$\begin{aligned} & \sum_{k=1}^K \sum_{l=1}^K \phi_{kl} \left(\Psi(\pi_{0l}) - \Psi(\pi_{0l} + \pi_{1l}) \right) + \sum_{k=1}^K \sum_{l=1}^K \phi_{kl} \sum_{m=1}^{l-1} \left(\Psi(\pi_{1m}) - \Psi(\pi_{0m} + \pi_{1m}) \right) \\ & + (\kappa - 1) (\Psi(\pi_{1l}) - \Psi(\pi_{0l} + \pi_{1l})) \\ & + \log \Gamma(\pi_{0l}) + \log \Gamma(\pi_{1l}) - \log \Gamma(\pi_{0l} + \pi_{1l}) \\ & - (\pi_{0l} - 1) \Psi(\pi_{0l}) - (\pi_{1l} - 1) \Psi(\pi_{1l}) + (\pi_{0l} + \pi_{1l} - 2) \Psi(\pi_{0l} + \pi_{1l}) \end{aligned}$$

Noting that $\frac{d}{dx} \log \Gamma(x) = \Psi(x)$, the derivative with respect to $\pi_{0,l}$ is :

$$\begin{aligned} & \sum_{k=1}^K \phi_{kl} \left(\Psi'(\pi_{0l}) - \Psi'(\pi_{0l} + \pi_{1l}) \right) + \sum_{k=1}^K \sum_{m>l} \phi_{km} \left(-\Psi'(\pi_{0l} + \pi_{1l}) \right) \\ & + (\kappa - 1) (-\Psi'(\pi_{0l} + \pi_{1l})) + \Psi(\pi_{0l}) - \Psi(\pi_{0l} + \pi_{1l}) - (\pi_{0l} - 1) \Psi'(\pi_{0l}) - \Psi(\pi_{0l}) \\ & + (\pi_{0l} + \pi_{1l} - 2) \Psi'(\pi_{0l} + \pi_{1l}) + \Psi(\pi_{0l} + \pi_{1l}) \end{aligned}$$

Grouping terms and applying cancellation when possible we have:

$$\Psi'(\pi_{0l}) \left[\sum_{k=1}^K \phi_{kl} - \pi_{0l} + 1 \right] + \Psi'(\pi_{0l} + \pi_{1l}) \left[(\pi_{0l} + \pi_{1l} - 1 - \kappa) - \sum_{k=1}^K \phi_{kl} - \sum_{k=1}^K \sum_{m>l} \phi_{km} \right]$$

The derivative with respect to π_{1l} is:

$$\begin{aligned} & \sum_{k=1}^K \phi_{kl} \left(-\Psi'(\pi_{0l} + \pi_{1l}) \right) + \sum_{k=1}^K \sum_{m>l} \phi_{km} \left(\Psi'(\pi_{1l}) - \Psi'(\pi_{0l} + \pi_{1l}) \right) \\ & + (\kappa - 1)(\Psi'(\pi_{1l}) - \Psi'(\pi_{0l} + \pi_{1l})) + \Psi(\pi_{1l}) - \Psi(\pi_{0l} + \pi_{1l}) - (\pi_{1l} - 1)\Psi'(\pi_{1l}) - \Psi(\pi_{1l}) \\ & + (\pi_{0l} + \pi_{1l} - 2)\Psi'(\pi_{0l} + \pi_{1l}) + \Psi(\pi_{0l} + \pi_{1l}) \end{aligned}$$

Grouping terms and applying cancellation:

$$\Psi'(\pi_{1l}) \left[\sum_{k=1}^K \sum_{m>l} \phi_{km} - \pi_{1l} + \kappa \right] + \Psi'(\pi_{0l} + \pi_{1l}) \left[(\pi_{0l} + \pi_{1l} - 1 - \kappa) - \sum_{k=1}^K \phi_{kl} - \sum_{k=1}^K \sum_{m>l} \phi_{km} \right]$$

Hence we see that the updates are:

$$\begin{aligned} \pi_{0,l} &= 1 + \sum_{k=1}^K \phi_{kl} \\ \pi_{1,l} &= \kappa + \sum_{k=1}^K \sum_{m>l} \phi_{k,l} \end{aligned}$$

2.6 Updates for $q_{i,a}$ and $q_{i,b}$

Under the correlated prior model described in Section 3.3., with auxilliary latent variable $p_i \sim \text{Beta}(\rho, \rho)$ and $A_{ik}|p_i \sim \text{Bernoulli}(p_i)$. Taking the variational posterior to be $p_i \sim \text{Beta}(q_{i,a}, q_{i,b})$, and applying the generalized exponential family update rule in Blei et al. (2017) (Equations 40, 47) gives the updates:

$$q_{i,a} = \rho + \sum_{k=1}^K \alpha_{ik}$$

$$q_{i,b} = \rho + K - \sum_{k=1}^K \alpha_{ik}$$

Finally expectation of the $\log p(A_{ik}|p_i)$ term is:

$$\begin{aligned} \mathbb{E} \log p(A_{ik}|p_i) &= \mathbb{E} A_{ik} \log p_i + (1 - A_{ik}) \log(1 - p_i) \\ &= \alpha_{ik} (\Psi(q_{i,a}) - \Psi(q_{i,a} + q_{i,b})) + (1 - \alpha_{ik}) (\Psi(q_{i,b}) - \Psi(q_{i,a} + q_{i,b})) \\ &= \alpha_{ik} (\Psi(q_{i,a}) - \Psi(q_{i,b})) + C \end{aligned}$$

where C is constant with respect to α_{ik} . Hence, the update for α_{ik} has an additional term of $\Psi(q_{i,a}) - \Psi(q_{i,b})$, i.e. we have:

$$\alpha_{ik} \leftarrow \sigma \left(\nabla_{\alpha_{ik}} \tilde{L}(\alpha, \gamma, \phi, \pi_0, \pi_1) + \Psi(q_{i,a}) - \Psi(q_{i,b}) \right)$$

3 Experimental details

Optimization. For all implementations, we defined convergence as the evidence lower bound increasing by less than a tolerance of $1e - 9$. Furthermore, we randomize each sequence of coordinate-wise updates, using a permutation selected uniformly at random. For the initialization strategy, we estimate G via Algorithm 21.1 of Barber (2012), which is implemented in the package `sklearn.decomposition.FactorAnalysis`. In all experiments we set the number of components to be P and the rotation to be varimax. Let the $P \times K_{\max}$ components matrix resulting from the factor analysis procedure be G_0 . We choose our initialization for G as $\text{softmax}_{\tau}(G_0)$ where the softmax is applied elementwise and τ is the temperature parameter. In all experiments we choose τ as $\max(G_0)/10$. For initialization of α , s and g , we iterate the updates (after initializing $\alpha_{ik} \sim \text{Unif}[0, 1]$, $s_j \sim \text{Unif}[0, 0.5]$ and $g_j \sim \text{Unif}[0, 0.5]$)

$$\begin{aligned}
\alpha_{ik} &\leftarrow \sigma\left(\nabla_{\alpha_{ik}} \tilde{L}(\alpha, \gamma, \phi, \pi_0, \pi_1)\right) \\
s_j &\leftarrow \frac{\sum_{i=1}^N (1 - Y_{ij}) \left(\prod_{k=1}^K (1 - (1 - \alpha_{ik})\gamma_{jk})\right)}{\sum_{i=1}^N \prod_{k=1}^K (1 - (1 - \alpha_{ik})\gamma_{jk})} \\
g_j &\leftarrow \frac{\sum_{i=1}^N Y_{ij} (1 - \prod_{k=1}^K (1 - (1 - \alpha_{ik})\gamma_{jk}))}{\sum_{i=1}^N (1 - \prod_{k=1}^K (1 - (1 - \alpha_{ik})\gamma_{jk}))}
\end{aligned}$$

until convergence, with γ fixed to $\text{softmax}_\tau(G_0)$. With this initialization for (α, γ, s, g) , we initialize the remaining parameters randomly. Namely:

$$\kappa_l \sim \text{Unif}[\Delta^{K_{\max}-1}]$$

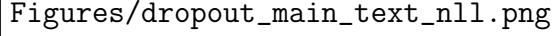
$$\pi_{l0} \sim \text{Unif}[1, 1.1]$$

$$\pi_{l1} \sim \text{Unif}[1, 1.1]$$

for all $l \in [K_{\max}]$, where $\Delta^{K_{\max}-1}$ is the $K_{\max}$ -simplex. We then fit the algorithm in two stages. We first run the Algorithm 1 updates for α , γ , s and g until convergence. Then we run all Algorithm 1 updates until convergence.

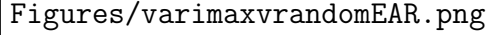
TIMSS 2011 Study. We downloaded the TIMSS 2011 Austria study from the R package CDM, specifically under `data.timss11.G4.AUT`. This included binary observation matrix as well as a $P \times 9$ expert Q matrix. Because not all pairs of students and questions are observed, we modify the evidence lower bound to only sum over the observed set of students and questions. The expert Q matrix is padded with 0s so that it is of dimension $P \times K_{\max}$. This padded expert Q matrix replaces $\text{softmax}_\tau(G_0)$ of the previous section, and all remaining details are the same.

Bone Marrow scATACseq Study. Count matrices and cell type annotations were downloaded from <https://zenodo.org/records/6383269>. We determined cell-type informative



Figures/dropout_main_text_nll.png

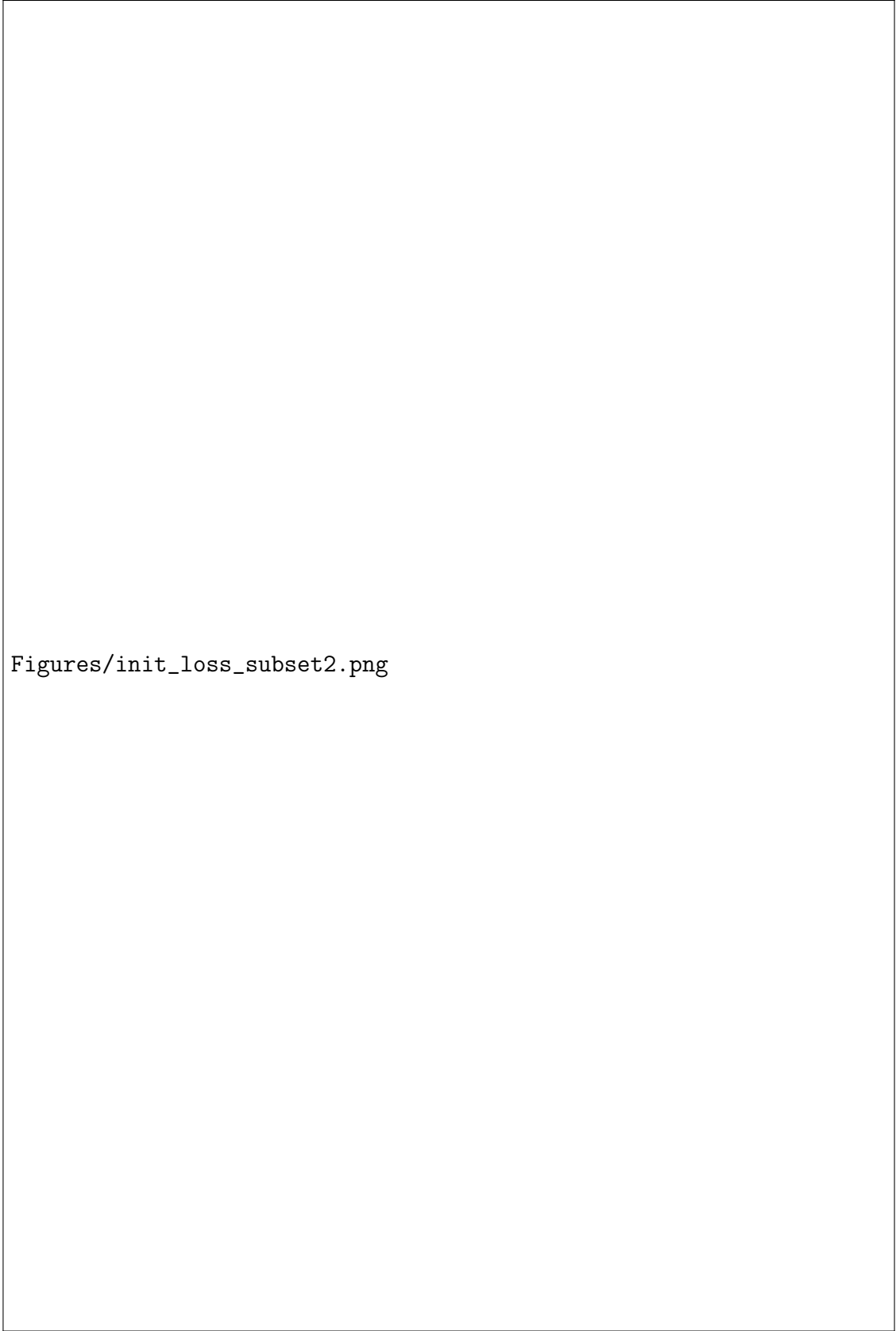
Supp. Fig 1: *Dropout improves algorithm performance.* Negative log likelihood (NLL) across a number of experimental settings and models evaluated at the VI posterior mode with dropout and without dropout. Here, we evaluate the base variational parameterization with uniform random initialization, with $\rho = 0, s = 0.2, g = 0.2$.



Figures/varimaxvrandomEAR.png

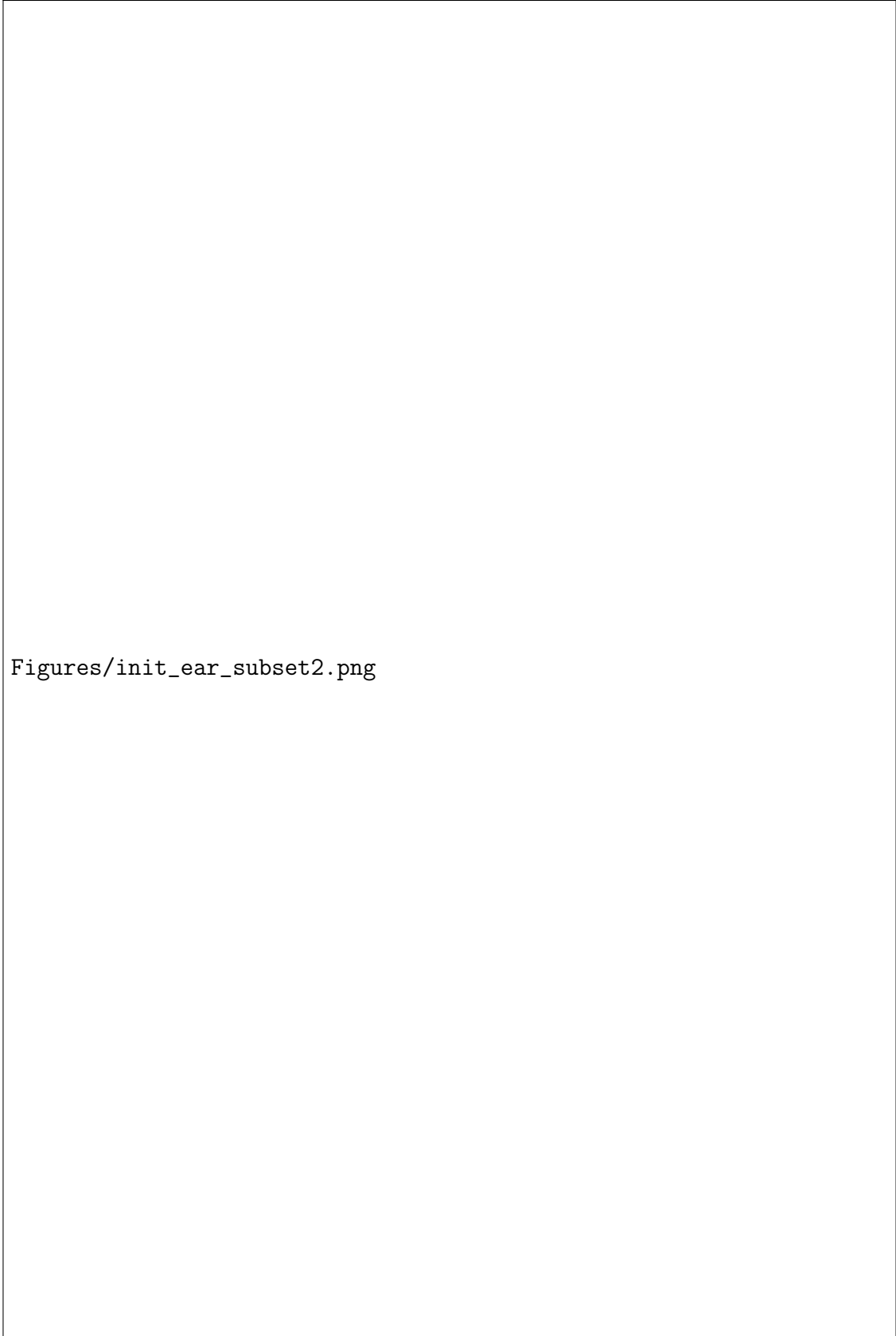
Supp. Fig 2: *Varimax initialization without dropout is superior to uniform random initialization with dropout.* Elementwise accuracy rate (EAR) across a number of experimental settings evaluated at the VI posterior mode with varimax initialization or with random uniform initialization with dropout. Here, we evaluate the base variational parameterization with $\rho = 0, s = 0.2, g = 0.2$.

peaks as follows. For each cell type and each peak, a Wilcoxon rank sum test was applied between that cell type and the remaining cell types. The peaks with the 16 smallest p values were retained. Combining across the 8 cell types, this led to 122 informative peaks (122 instead of 128 because some peaks were shared in multiple cell types). The count matrix was then binarized based on whether the number of counts is greater than 0, leading to a binary 6881×122 matrix. 225 cells were dropped based on having less than 1.5% of the peaks, leading to a binary 6656×122 matrix. We then ran Algorithm 1 with $K_{\max}$ fixed to 35. We then smoothed estimates α_{ik} by the average $\frac{1}{15} \sum_{j \in \mathcal{N}(i)} \alpha_{jk}$ where $\mathcal{N}(i)$ contains the indices of the 15 (inclusive) nearest neighbors of cell i .



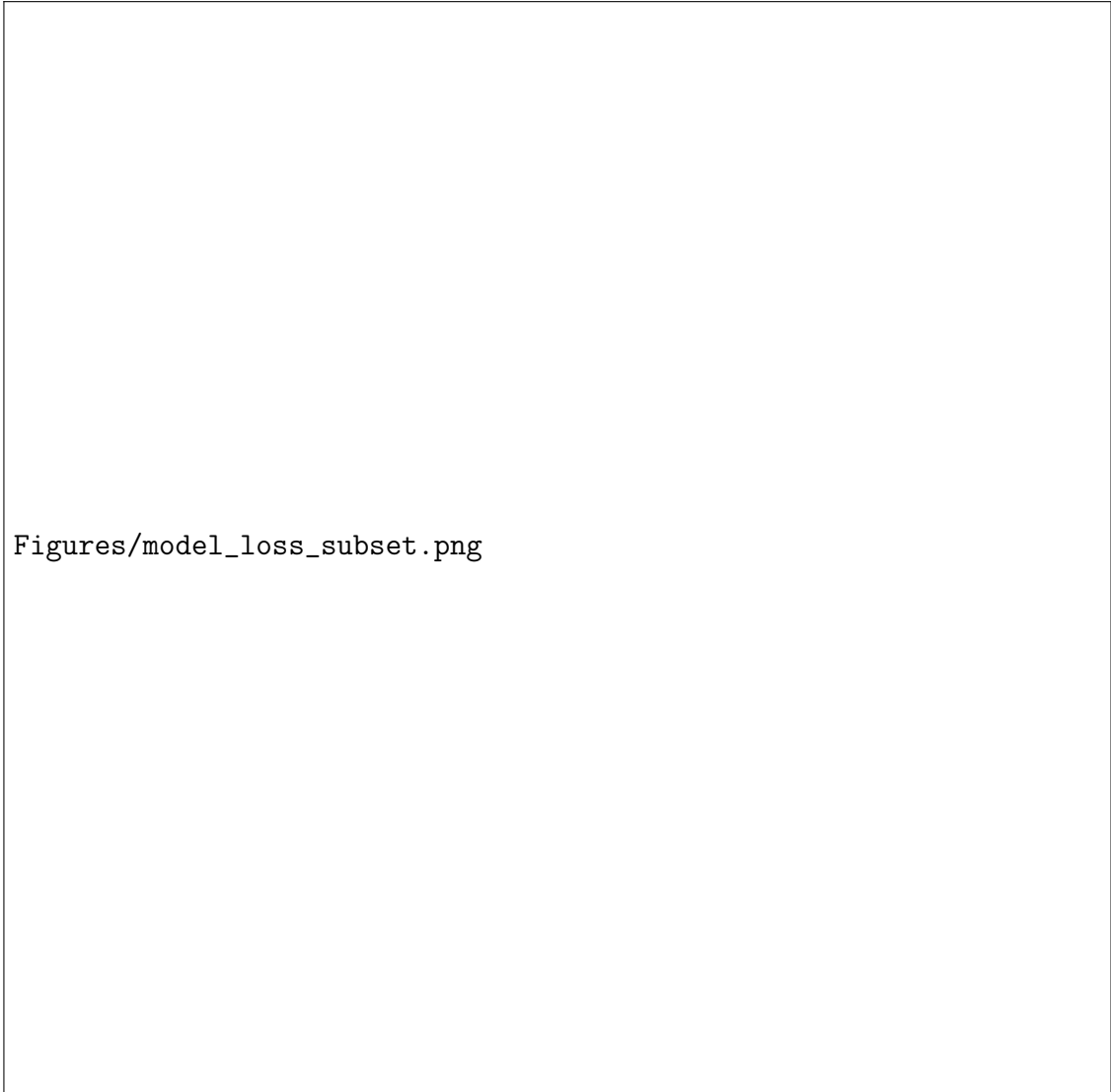
Figures/init_loss_subset2.png

Supp. Fig 3: *Initialization with varimax improves NLL across a wide array of experimental settings.* DINA negative log likelihood (NLL) across a number of experimental settings and models for both uniform random initialization and initialization via the factor analysis with varimax.



Figures/init_ear_subset2.png

Supp. Fig 4: *Initialization with varimax improves EAR across a wide array of experimental settings.* EAR across a number of experimental settings and models for both uniform random initialization and initialization via the factor analysis with varimax.



Figures/model_loss_subset.png

Supp. Fig 5: DINA negative log likelihood (NLL) across a number of experimental settings for the base DINA-CSP model, the DINA-CSP model with correlated prior and the DINA-CSP model with full parameterization of the latent skills A_i .



Figures/model_ear_subset.png

Supp. Fig 6: EAR values across a number of experimental settings for the base DINA-CSP model, the DINA-CSP model with correlated prior and the DINA-CSP model with full parameterization of the latent skills A_i .

4 Previous Theorem Proofs

Lemma 6 (Error bound for α_{ik}). *Let $\text{err}(\alpha_{ik}) = |\alpha_{ik} - A_{ik}|$. Assume that $\rho := \log(\frac{1-s_j}{g_j}) = \log(\frac{1-g_j}{s_j}) > 0$ with $\rho < \infty$ (This can be relaxed to allowing s_j and g_j to vary in a range).*

Consider the event

$$\mathcal{E} = \left\{ (2A_{ik} - 1) \left(\sum_{j=1}^P G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \times \rho(2Y_{ij} - 1) \right) > Pc \right\} \quad (19)$$

for a constant $c > 0$. For suitably chosen c , this event will occur with high probability. Then if $\mathcal{E}$ holds and we have an initialization such that $|\alpha_{il} - A_{il}| < \varrho$, $|\gamma_{jl} - G_{jl}| < \varrho$ for all (i, j, l) , after one CAVI update

$$\text{err}(\alpha_{ik}) \leq \exp \left(-P(c - \rho(1 - (1 - \varrho)^K)) \right)$$

Proof. The update for α_{ik} is given by:

$$\begin{aligned} \hat{\alpha}_{ik} &= \sigma(\nabla_{\alpha_{ik}} \tilde{L}(\alpha, \gamma)) \\ &= \sigma \left(\sum_{j=1}^P [Y_{ij} \log(\frac{1-s_j}{g_j}) - (1 - Y_{ij}) \log(\frac{1-g_j}{s_j})] \times \gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \right) \\ &= \sigma \left(\sum_{j=1}^P \rho(2Y_{ij} - 1) \times \gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \right) \\ &= \sigma \left(\sum_{j=1}^P \rho(2Y_{ij} - 1) \times \left(G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) + \Delta_j \right) \right) \end{aligned}$$

where

$$\Delta_j := \gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) - G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl})$$

We now establish the bound $|\Delta_j| \leq 1 - (1 - \varrho)^K$ by casework. Suppose $G_{jk} = 0$. Then by

assumption $|\gamma_{jk}| \leq \varrho$. We have:

$$|\Delta_j| = \gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \leq \gamma_{jk} \leq \varrho$$

Otherwise suppose $\prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = 0$. Then there is at least one index m such that $A_{im} = 0$ and $G_{jm} = 1$. Then, by assumption, $\alpha_{im} \leq \varrho$ and $\gamma_{jm} \geq 1 - \varrho$. Then, combining these results:

$$(1 - \alpha_{im})\gamma_{jm} \geq (1 - \varrho)^2$$

$$|\Delta_j| = \gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \leq 1 - (1 - \alpha_{im})\gamma_{jm} \leq 1 - (1 - \varrho)^2$$

The final case has $G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) = 1$, which implies $G_{jk} = 1$, and in turn $\gamma_{jk} \geq 1 - \varrho$. Further, if $G_{jl} = 0$ then $\gamma_{jl} \leq \varrho$, which implies $(1 - \alpha_{jl})\gamma_{jl} \leq \gamma_{jl} \leq \varrho$. This establishes $1 - (1 - \alpha_{jl})\gamma_{jl} \geq 1 - \varrho$. If $G_{jl} = 1$, then $A_{jl} = 1$ and

$$\alpha_{jl} \geq 1 - \varrho$$

$$1 - \alpha_{jl} \leq \varrho$$

$$(1 - \alpha_{jl})\gamma_{jl} \leq \varrho$$

Together this establishes each term of $\gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl})$ is greater than or equal to $1 - \varrho$, and hence:

$$\gamma_{jk} \prod_{l \neq k} (1 - (1 - \alpha_{il})\gamma_{jl}) \geq (1 - \varrho)^K$$

Combining the results for all possible cases we have:

$$|\Delta_j| \leq \max(\varrho, 1 - (1 - \varrho)^2, 1 - (1 - \varrho)^K)$$

For $K \geq 2$, we can combine these as $|\Delta_j| \leq 1 - (1 - \varrho)^K$.

To establish the result, we proceed by casework on A_{ik} . For $A_{ik} = 0$, we have:

$$\begin{aligned} \text{err}(\alpha_{ik}) &\leq \exp \left(\sum_{j=1}^P \rho(2Y_{ij} - 1) \times \left(G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) + \Delta_j \right) \right) \\ &\leq \exp \left(-Pc + \rho \sum_{j=1}^P (2Y_{ij} - 1)(1 - (1 - \varrho)^K) \right) \\ &\leq \exp \left(-P(c - \rho(1 - (1 - \varrho)^K)) \right) \end{aligned}$$

where we use the fact that event $\mathcal{E}$ occurs. For $A_{ik} = 1$,

$$\begin{aligned} \text{err}(\alpha_{ik}) &\leq \exp \left(- \sum_{j=1}^P \rho(2Y_{ij} - 1) \times \left(G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) + \Delta_j \right) \right) \\ &\leq \exp \left(-Pc + \rho \sum_{j=1}^P (2Y_{ij} - 1)(1 - (1 - \varrho)^K) \right) \\ &\leq \exp \left(-P(c - \rho(1 - (1 - \varrho)^K)) \right) \end{aligned}$$

where we have used the event $\mathcal{E}$ and $-\rho(2Y_{ij} - 1)\Delta_j \leq \rho(1 - (1 - \varrho)^K)$. $\square$

Lemma 7 (Concentration of event $\mathcal{E}$). *We require the identifying assumptions $s < 1/2 - \epsilon$, $g < 1/2 - \epsilon$ and Assumption 1. We further identify the constant c in Lemma 1. Namely, let $c := \rho \times \epsilon \times \Xi$ (Ξ as defined in Assumption 1, and ρ defined in Lemma 1). Then*

$$\mathbb{P}[\mathcal{E}] \geq 1 - NK \exp \left(\frac{-P \times c}{(\frac{2}{3} + \frac{2}{\epsilon})\rho} \right)$$

Proof. We assume the case $A_{ik} = 1$. Under the assumptions, the case $A_{ik} = 0$ can be shown to be equivalent. Define:

$$u_j := G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \times \rho(2Y_{ij} - 1)$$

Note that these variables are bounded so we aim to apply Bernstein's inequality. We first compute the expectation,

$$\mathbb{E}[u_j] = G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \times 2\rho\left(\frac{1}{2} - s\right)$$

And define:

$$\begin{aligned} D &:= \mathbb{E}\left[\sum_{j=1}^P u_j\right] = 2\rho\left(\frac{1}{2} - s\right) \sum_{j=1}^P G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \\ &\geq 2\rho \times \epsilon \Xi P = 2Pc \end{aligned}$$

where the second line follows from the assumptions. Note that $u_j \in \{-\rho, \rho, 0\}$ almost surely.

The second moment can thus be computed directly as:

$$\begin{aligned} \mathbb{E}[u_j^2] &= \rho^2 \times G_{jk} \prod_{l \neq k} (1 - (1 - A_{il})G_{jl}) \\ \sum_{j=1}^P \mathbb{E}[u_j^2] &= \frac{\rho D}{2(\frac{1}{2} - s)} \end{aligned}$$

Applying Bernstein's inequality then gives (note that the inequality holds for non-centered random variables):

$$\mathbb{P}\left(\sum_{j=1}^P u_j \geq \frac{D}{2}\right) \geq 1 - \exp\left(-\frac{\frac{1}{2}\left(\frac{D}{2}\right)^2}{\frac{D\rho}{2(\frac{1}{2}-s)} + \frac{1}{6}\rho D}\right)$$

$$\begin{aligned}
&\geq 1 - \exp\left(-\frac{D}{(\frac{4}{3} + \frac{4}{(\frac{1}{2}-s)})\rho}\right) \\
&\geq 1 - \exp\left(-\frac{D}{(\frac{4}{3} + \frac{4}{\epsilon})\rho}\right) \\
&\geq 1 - \exp\left(-\frac{2Pc}{(\frac{4}{3} + \frac{4}{\epsilon})\rho}\right)
\end{aligned}$$

Finally we note $D/2 \geq Pc$, so

$$\mathbb{P}\left(\sum_{j=1}^P u_j \geq Pc\right) \geq 1 - \exp\left(-\frac{Pc}{(\frac{2}{3} + \frac{2}{\epsilon})\rho}\right)$$

Applying the union bound shows the desired result. □

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